

VIETNAM NATIONAL UNIVERSITY, HANOI
INTERNATIONAL SCHOOL



STUDENT RESEARCH REPORT

**TRAINING SELF-AWARE LLMS AGAINST PROMPT INJECTION
VIA PREFERENCE OPTIMIZATION**

CN.NC.SV.25_08

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Thank you all for helping us achieve our goals!

INTRODUCTION

1. Project Name

English.

Training Self-Aware LLMs Against Prompt Injection via Preference Optimization

Vietnamese.

Huấn Luyện LLMs Tự Nhận Thức Thông Qua Tối Ưu Hóa Ưu Tiên

2. Project Code

CN.NC.SV.25_08

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5. Abstract

English. Large language models deployed in agentic applications face a critical and unresolved architectural vulnerability: prompt injection attacks exploit the model's fundamental inability to distinguish trusted instructions from untrusted external data within the same token sequence. Ranked the foremost security risk for LLM-integrated systems by the OWASP Top 10 for LLM Applications 2025, this threat is both

practically severe and academically underexplored in non-English contexts. Existing model-level defenses, most notably SecAlign++, the leading open-source preference optimization framework, have been developed and validated exclusively on English-language data, leaving low-resource languages such as Vietnamese entirely without coverage. This study addresses that gap along four fronts. First, we construct a 26,819-sample bilingual Vietnamese-English preference dataset combining five curated sources, with Vietnamese content comprising 52.2% of training samples and representing the first resource of its kind designed specifically for prompt injection defense. Second, we adapt the SecAlign++ fine-tuning pipeline for Llama-3.1-8B-Instruct using Direct Preference Optimization augmented with Rejection Pair Optimization and label smoothing, reducing the average attack success rate from 91.75% to 74.00% while preserving general utility with a negligible degradation of 0.03 percentage points on the Massive Multitask Language Understanding benchmark. Third, we systematically map SecAlign++ attack mechanisms to the OWASP Top 10 risk framework, establishing a standardized baseline for Vietnamese AI security assessment. Fourth, through a code-level gap analysis of the original codebase, we identify and empirically validate ten novel attack vectors absent from the original training protocol, among them encoding obfuscation, cross-lingual injection, and multi-turn gradual escalation, and demonstrate that 2,000 targeted preference samples produce measurable defense improvements across all ten. A cross-architecture adaptation recipe for GPT-style models is additionally provided. Taken together, these contributions establish the first open-source, model-level prompt injection defense baseline for the Vietnamese language and offer a reusable methodology for extending preference-optimization-based security to other low-resource linguistic contexts.

Vietnamese. Các mô hình ngôn ngữ lớn (LLM) triển khai trong ứng dụng tác vụ tự động đang đối mặt với một lỗ hổng kiến trúc nghiêm trọng chưa được giải quyết: tấn công prompt injection khai thác điểm yếu cốt lõi khi mô hình không thể phân biệt giữa chỉ dẫn tin cậy và dữ liệu bên ngoài không tin cậy trong cùng chuỗi token. Được OWASP Top 10 for LLM Applications 2025 xếp hạng là rủi ro an ninh hàng đầu đối với hệ thống tích hợp LLM, mối đe dọa này vừa có tính thực tiễn nghiêm trọng vừa ít

được nghiên cứu trong các ngữ cảnh phi tiếng Anh. Các biện pháp phòng thủ ở mức mô hình hiện có, điển hình là SecAlign++, framework tối ưu hóa sở ưu tiên mã nguồn mở hàng đầu, chỉ được phát triển và kiểm chứng trên dữ liệu tiếng Anh, khiến các ngôn ngữ ít tài nguyên như tiếng Việt hoàn toàn chưa được bao phủ. Nghiên cứu này giải quyết khoảng trống đó trên bốn phương diện. Thứ nhất, chúng tôi xây dựng bộ dữ liệu song ngữ Việt-Anh gồm 26.819 mẫu từ năm nguồn tuyển chọn, trong đó nội dung tiếng Việt chiếm 52,2% mẫu huấn luyện, trở thành tài nguyên đầu tiên được thiết kế riêng cho phòng thủ prompt injection. Thứ hai, chúng tôi điều chỉnh pipeline tinh chỉnh SecAlign++ cho Llama-3.1-8B-Instruct bằng Tối ưu Ưu tiên Trực tiếp (Directed Preference Optimization - DPO) kết hợp Tối ưu Cặp Từ chối (Rejection Pair Optimization) và làm mượt nhãn, giúp giảm tỷ lệ tấn công thành công trung bình từ 91,75% xuống 74,00% đồng thời duy trì tính hữu dụng tổng thể với suy giảm không đáng kể 0,03 điểm phần trăm trên benchmark Massive Multitask Language Understanding. Thứ ba, chúng tôi hệ thống hóa cơ chế tấn công của SecAlign++ theo khung rủi ro OWASP Top 10, thiết lập chuẩn đánh giá an ninh AI cho tiếng Việt. Thứ tư, thông qua phân tích khoảng trống ở mức mã nguồn, chúng tôi xác định và kiểm chứng thực nghiệm mười hướng tấn công mới chưa có trong giao thức huấn luyện gốc, bao gồm che giấu mã hóa, tiêm nhiễm xuyên ngôn ngữ, và leo thang dần theo nhiều lượt, đồng thời chứng minh rằng 2.000 mẫu ưu tiên mục tiêu tạo ra cải thiện phòng thủ đo lường được trên cả mười vector. Ngoài ra, nghiên cứu cung cấp công thức thích ứng chéo cho các mô hình kiểu GPT. Tổng thể, các đóng góp này thiết lập baseline phòng thủ prompt injection đầu tiên ở mức mô hình cho tiếng Việt và đưa ra phương pháp luận có thể tái sử dụng để mở rộng bảo mật dựa trên tối ưu hóa sở thích sang các ngữ cảnh ngôn ngữ ít tài nguyên khác.

6. Keywords

English. LLM Security · Self-Aware Language Model · Vietnamese Natural Language Processing.

Vietnamese. Bảo mật mô hình ngôn ngữ lớn · Mô hình ngôn ngữ tự nhận thức · Xử lý ngôn ngữ tự nhiên tiếng Việt.

SUMMARY REPORT IN STUDENT RESEARCH,

2025-2026 ACADEMIC YEAR

The research report is organized into five main chapters, along with additional sections for Acknowledgments, Abbreviations, and References.

Chapter 1: Literature Review

This chapter examines security in large language models, focusing on prompt injection as a significant threat. It identifies defense gaps for low-resource languages, such as Vietnamese, and outlines the study's objectives.

Chapter 2: Data & Methodology

This section details the bilingual Vietnamese-English dataset and the preprocessing steps involved. It also introduces the adapted SecAlign++ framework for fine-tuning the Llama-3.1-8B-Instruct model.

Chapter 3: Results & Discussions

Here, the experimental results are presented. The chapter assesses the effectiveness of the defense mechanism against prompt injection and evaluates the trade-off between utility and security.

Chapter 4: Conclusion & Recommendations

This chapter summarizes the key contributions, evaluates the study's objectives, and provides recommendations for future research.

Chapter 5: Appendix

The appendix includes supplementary materials that ensure reproducibility and support further research.

Acknowledgments

This section recognizes the support and contributions of individuals and organizations that made this research possible.

Lastly, the report features dedicated sections for **Abbreviations** and **References** to ensure that all terms and cited works are clearly defined and properly documented.

Chapter 1. Literature Review

1.1 Necessity

1.1.1 Background of the Study

Over the past decade, large language models (LLMs) have undergone a significant transformation in their role within software systems. Initially, LLMs were deployed primarily as conversational interfaces, functioning as standalone systems that processed text queries and returned responses within a limited context window. However, contemporary uses are markedly different. LLMs now operate as orchestrators within agentic AI architectures, effectively linking external tools, application programming interfaces, databases, and real-time data sources to autonomously execute complex, multi-step tasks on behalf of users. These LLM-integrated applications enable a diverse range of capabilities, from autonomous document analysis and web navigation to multi-turn task planning and tool-calling workflows. Real-world implementations have been successfully demonstrated across various production platforms, including Google Bard, Slack AI, Microsoft Copilot, Anthropic Claude, and OpenAI Operator (*S. Chen et al., 2026*).

This shift from isolated chatbots to agentic orchestrators creates a fundamentally new attack surface that conventional cybersecurity mechanisms are ill-equipped to handle. In traditional software systems, the separation between executable code and user-supplied data is maintained at the architectural level through memory protection, privilege rings, and input sanitization processes. In contrast, LLMs treat all inputs (including system instructions, user queries, and externally retrieved data) as indistinguishable token sequences processed by the same self-attention mechanism. There are no hardware-level or architectural barriers that differentiate a legitimate system command from user-controlled or externally fetched text (*Wallace et al., 2024*). Consequently, an adversary who can manipulate the data retrieved by an LLM-integrated application could, in theory, redirect the model's behavior without ever altering the system prompt or the user interface directly.

1.1.2 Statement of the Problem

The core vulnerability addressed in this research is prompt injection (PI), which refers to the systematic difficulty that language models (LLMs) face in distinguishing between trusted instructions and untrusted data presented within the same input context (*S. Chen et al., 2025*). This issue is not a software defect that can be rectified through conventional means; rather, it stems from the structural design of Transformer-based models, which process input as a continuous sequence of tokens. In this context, salience is determined by learned statistical relationships rather than any notion of provenance or trust level (*Wallace et al., 2024*). Consequently, malicious instructions embedded in externally sourced content (such as a webpage, an API response, or a document) can override legitimate system instructions, leading the model to exfiltrate sensitive data, execute unauthorized actions, or produce outputs that serve the attacker rather than the system designer.

The threat is both acknowledged and unresolved at scale. According to the OWASP Top 10 for LLM Applications 2025, prompt injection stands as the foremost security risk for LLM-integrated systems (*OWASP GenAI Security Project, 2024*). Successful attacks have been demonstrated against production systems across various sectors, and the risk escalates with the model's capability: more advanced instruction-following models are empirically more vulnerable when left unprotected (*S. Chen et al., 2026*). Existing model-level defenses, which embed a security policy directly into model weights through fine-tuning, face three systemic limitations that this study aims to address:

- 1) **Closed-source Deployment:** The most robust model-level defenses are integrated into proprietary commercial APIs, leaving the research community without access to a public training methodology, evaluation protocols, or reproducible baselines.
- 2) **English-only Validation:** All currently published model-level defenses have been developed and evaluated solely using English-language datasets. There has been no empirical investigation into whether the security policies learned through fine-tuning can generalize across different languages.
- 3) **Absence of Vietnamese Coverage:** Vietnamese is an active and growing language for LLM-integrated applications within Vietnam's administrative,

educational, and healthcare sectors. Currently, there are no open-source model-level prompt injection defenses available for Vietnamese, nor has any dataset with diverse attack coverage in Vietnamese prompt injection been published.

1.1.3 Purpose and Significance of the Study

This study explores the intersection of LLM security and Vietnamese natural language processing by adapting SecAlign++, the leading model-level defense framework (*S. Chen et al., 2026*), for bilingual deployment in Vietnamese and English. The significance of this research operates at three levels.

At the scientific level, this represents the first known effort to extend a preference optimization-based prompt injection defense to a low-resource language using a bilingual training dataset. It provides new empirical evidence on whether the security policy encoded through direct preference optimization (*DPO; Rafailov et al., 2023*) is transferable across different linguistic environments.

At the practical level, the study introduces two reusable open-source artifacts: a fine-tuned Llama-3.1-8B-Instruct model that incorporates bilingual prompt injection defense, and a 26,819 - sample Vietnamese-English preference dataset developed for AI safety research.

At the community level, this work establishes the first baseline for Vietnamese prompt injection defense, aligns attack mechanisms with the OWASP Top 10 for LLM 2025 risk framework, and proposes a taxonomy of ten novel attack vectors not included in the original SecAlign++ training protocol. This creates a structured foundation for future security research in the Vietnamese language context.

1.2 Overview

1.2.1 Overview of the Research Domain

Defenses against prompt injection can be categorized along a primary axis: whether security measures are implemented at the system level, externally to the model, or at the model level, incorporated within the model's weights and inference behavior. This distinction has significant implications for the robustness that can be achieved.

System-level defenses operate as intermediaries between the large language model (LLM) and its inputs. These defenses may detect injections before the model processes them; examples include Meta PromptGuard, Azure Prompt Shields, and LlamaFirewall. They can also employ in-context prompting techniques, such as spotlighting or sandwich prompting, or filter retrieved data to eliminate injection candidates prior to their inclusion in the context window. Notably, system-level defenses can be applied without modifying the underlying model and can be layered onto existing deployments. However, a fundamental limitation is that the model itself remains vulnerable; a sufficiently adaptive attacker may craft injections that bypass detection pipelines while still being effective against an unprotected model (*S. Chen et al., 2025*).

In contrast, model-level defenses address the issue at its core by modifying the model's internal behavior through fine-tuning. These models are trained to regard instructions from untrusted data sources as non-executable content, regardless of their surface form or linguistic framing. Since this defense is embedded within the model weights rather than relying on an external filter, it cannot be circumvented by attacks that avoid input preprocessing. Empirical studies consistently indicate that model-level defenses outperform system-level approaches in the face of both optimization-free and optimization-based attacks. This research specifically focuses on model-level defenses, particularly the preference optimization variant.

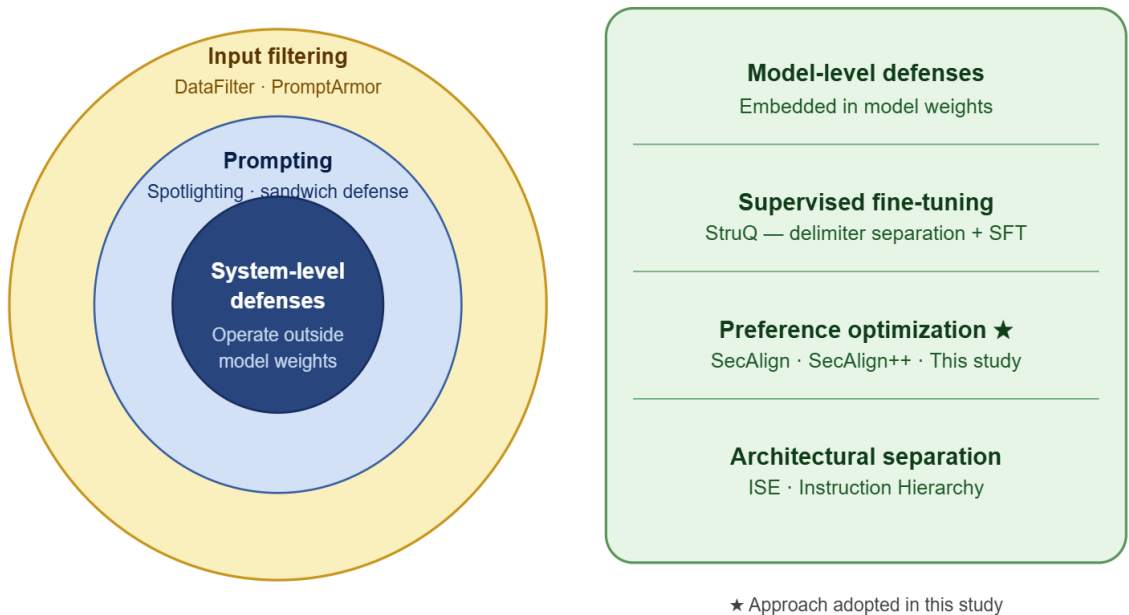


Figure 1.1. Taxonomy of prompt injection defense strategies, with the approach adopted in this study indicated.

1.2.2 Related Work

The evolution of model-level defenses against prompt injection has followed a logical progression, wherein each subsequent generation directly addresses the limitations identified in earlier approaches.

- ❖ **StruQ** (*S. Chen et al., 2024*) introduced the foundational principle of structural separation by incorporating a new message role in the LLM's chat template. This addition explicitly distinguishes untrusted data from trusted instructions using special tokens, and the model is fine-tuned through supervised learning (SFT) to adhere to this separation. StruQ demonstrated strong protection against static, optimization-free attacks. However, it proved vulnerable to optimization-based adversaries: the Greedy Coordinate Gradient (GCG) attack (*Zou et al., 2023*) achieved attack success rates exceeding 50% against models defended by StruQ. This vulnerability arises because focusing solely on maximizing the likelihood of secure responses, without concurrently discouraging insecure ones, results in insufficient robustness.
- ❖ **SecAlign** (*S. Chen et al., 2025*) has reframed the challenge of prompt injection defense as a problem of preference optimization. When presented with a prompt-injected input, the model is trained using direct preference optimization (DPO; *Rafailov et al., 2023*) to favor responses that adhere to legitimate instructions over those that execute the injection. By explicitly working to minimize the likelihood of insecure outputs while maximizing the likelihood of secure ones, SecAlign achieves a significantly larger log-probability margin between these two response types; this characteristic is directly linked to empirically enhanced robustness. Notably, SecAlign is the first defense mechanism to lower the success rates of optimization-based attacks below 10% across all tested models while maintaining utility levels comparable to those of an undefended baseline.
- ❖ **Meta SecAlign / SecAlign++** (*S. Chen et al., 2026*) identifies two primary failure modes present in the original SecAlign and introduces targeted technical solutions to address them. Firstly, the practice of placing injections exclusively at the end of the data field during training leads the model to develop a positional shortcut; this results in the model disregarding the final instruction,

whether trusted or untrusted, instead of understanding the semantic principles crucial for prompt injection defense. To counter this, SecAlign++ employs a randomized injection positioning strategy: 45% of training injections are positioned at the beginning of the data field, 45% at the end, and 10% utilize the completion attack format. Secondly, the original SecAlign relied on low-quality, out-of-distribution ground-truth responses derived from an outdated annotator model for DPO training labels, thereby compromising both utility and security. SecAlign++ rectifies this issue by utilizing the outputs of the initialization model as training labels, ensuring consistency in distribution. These enhancements result in the first commercial-grade open-source prompt injection defense; META-SECALIGN-70B achieves an impressive 6.4% attack success rate on the SEP instruction-following benchmark and 1.9% on the AgentDojo agentic tool-calling benchmark, demonstrating competitiveness with proprietary systems, including GPT-5. This study successfully adapts the framework for the Vietnamese context.

- ❖ ***The Instruction Hierarchy*** (Wallace et al., 2024) introduces an alternative model-level approach that assigns explicit priority levels to different types of instructions: system prompts take precedence over user messages, which in turn take precedence over data or tool outputs. This model is designed to consistently prioritize higher-level instructions in situations where conflicts arise. When implemented in GPT-4o mini, this approach demonstrates strong resilience against optimization-free attacks. Notably, it does not utilize explicit supervision for undesirable responses, setting it apart architecturally from DPO-based methods.
- ❖ ***Instructional Segment Embedding (ISE)*** (Wu et al., 2024) presents an architectural framework that assigns distinct embedding offsets to various instruction segments. This design encodes trust levels at the representation level, moving beyond the reliance on token delimiters. ISE operates independently from preference optimization, and in principle, they can be integrated.

In the context of Vietnamese natural language processing (NLP), the PhoGPT model (Nguyen et al., 2023), boasting 4 billion parameters and pre-trained on a

102-billion-token Vietnamese corpus, stands as the benchmark among open-source language models tailored for Vietnamese. The ViHOS dataset (*Hoang et al., 2023*) comprises 11,056 human-annotated Vietnamese online comments that contain hateful and offensive content, developed specifically for hate speech detection. The accessibility of these resources lays the groundwork for the bilingual dataset created in this study, as neither has been utilized previously for prompt injection defense.

1.2.3 Research Gaps and Proposed Contributions

A systematic review of the literature reveals seven key research gaps. The first three focus on the lack of resources specific to the Vietnamese context, while the remaining four highlight limitations in the coverage of attack vectors within the SecAlign++ training framework.

Table 1.1. Identified research gaps

Gap ID	Description
GAP-01	There is no open-source model-level prompt injection defense available for Vietnamese.
GAP-02	All published training data for model-level defenses is exclusively in English; cross-lingual generalization has not been empirically explored.
GAP-03	A Vietnamese prompt injection dataset that encompasses a diverse range of attack types has yet to be published.
GAP-04	Encoding-based obfuscation attacks (such as Base64, ROT13, and leetspeak) are not included in the SecAlign++ training framework.
GAP-05	Current defenses do not address multi-turn progressive injection, where attack payloads are distributed across multiple conversation turns.
GAP-06	Context overflow and attention dilution attacks have not been systematically assessed.
GAP-07	There is no clearly defined defense mechanism against semantic injection without explicit override keywords.

This study makes four key contributions to address existing gaps:

C1 - Bilingual Preference Dataset (GAP-01, GAP-02, GAP-03). A preference dataset comprising 26,819 samples has been developed from five distinct sources: 10,813 English-language safety preference pairs obtained from PKU-SafeRLHF (*Ji et al., 2025*); 8,844 Vietnamese-language pairs sourced from the ViHOS hate speech

corpus (*Hoang et al., 2023*); 4,162 Vietnamese pairs from the ViHSD dataset; 1,000 self-constructed Vietnamese hallucination pairs; and 2,000 English-language pairs covering ten novel attack vectors. This dataset replaces the English-only baselines, with Vietnamese-language samples constituting approximately 52.2% of the complete dataset, significantly broadening the range of attack types.

C2 - Vietnamese-Adapted Llama-3.1-8B-Instruct (*GAP-01, GAP-02*). The fine-tuning pipeline of SecAlign++ has been adapted for Llama-3.1-8B-Instruct using the bilingual dataset. This adaptation includes expanded English injection templates, training data targeting Vietnamese hate speech, and a non-overlapping train/test split to ensure accurate measurement of generalization. Training is completed within 23 minutes on a single NVIDIA A100 GPU utilizing HuggingFace TRL. The study also investigates hallucination resistance on the synthetic subset and explores the utility-security trade-off behavior through LoRA α scaling analysis.

C3 - OWASP Mapping and Comprehensive Benchmarking (*GAP-01, GAP-02, GAP-03*). This section provides a systematic mapping of the attack mechanisms addressed by SecAlign++ to the OWASP Top 10 for LLM Applications 2025 (LLM01-LLM08; OWASP GenAI Security Project, 2024). This mapping establishes a standardized risk assessment framework suitable for AI deployments in Vietnam. Instead of relying on traditional, easily bypassed benchmarks, the evaluation encompasses a rigorous 10-Axis custom attack benchmark, alongside strictly targeted utility metrics: MMLU (0-shot), GSM8K, and False Refusal Rate (FRR). We successfully conducted a cross-model validation against the undefended Llama-3.1-8B-Instruct and the bilingual SeaLLM-7B-v2.5, proving the superiority of our defense.. Additionally, this work identifies and documents a previously unreported Base64 obfuscation vulnerability in the initial SecAlign++.

C4 - Novel Attack Taxonomy and Cross-Architecture Proposal (*GAP-04, GAP-05, GAP-06, GAP-07*). Through a code-level analysis of the SecAlign++ codebase, we propose ten novel attack vectors that were not included in the original training protocol. These vectors have been empirically verified on the undefended base model and incorporated into the preference training dataset. They include: encoding obfuscation, cross-lingual injection, keyword-free semantic injection, context overflow and attention dilution, Unicode homoglyph delimiter substitution, payload splitting,

multi-turn gradual escalation, conditional logic injection, Base64 obfuscation, and leetspeak substitution. Furthermore, we provide an adaptation recipe for applying SecAlign++ to GPT-style architectures, which addresses differences in chat template format, tokenizer configuration, and LoRA target module selection.

1.3 Objectives

1.3.1 General Objective

This study aims to develop and assess a Vietnamese LLM that is resistant to prompt-injection attacks through preference optimization, achieving performance that is competitive with the undefended baseline across both security and utility metrics, while making all components available as open-source resources.

1.3.2 Specific Objectives

Table 1.2. Specific objectives, contributions, and research gaps addressed

ID	Objective	Contri	Gaps
SO1	Construct a bilingual Vietnamese-English preference dataset of at least 10,000 samples with diverse attack type coverage	C1	GAP-01, 02, 03
SO2	Adapt the SecAlign++ framework for Llama-3.1-8B-Instruct using English injection templates and Vietnamese hate speech data; empirically validate hallucination resistance and LoRA α scaling behavior	C2	GAP-01, 02
SO3	Maintain utility and reasoning capabilities on standard benchmarks (MMLU 0-shot, GSM8K) while controlling the False Refusal Rate (FRR) compared to the base model.)	C2	GAP-01
SO4	Map SecAlign++ attack mechanisms to the OWASP Top 10 for LLM 2025 and conduct comprehensive benchmarking using a rigorous 10-Axis Custom Attack suite.	C3	GAP-01, 02, 03
SO5	Propose and empirically validate a novel attack taxonomy consisting of ten vectors, along with an adaptation recipe for SecAlign++ on GPT-style architectures	C4	GAP-04, 05, 06, 07

Chapter 2. Data & Methodology

2.1 Data

2.1.1 Dataset Overview and Design Principles

The original SecAlign++ framework utilizes the Cleaned-Alpaca corpus, an English-language instruction-tuning dataset comprising 19,157 samples, for training DPO, and AlpacaFarm, which includes 805 samples, serves as the evaluation benchmark to assess attack success rates and utility. However, neither of these datasets is applicable in this study due to computational constraints of the available training environment and the absence of Vietnamese content in the Cleaned-Alpaca corpus. This limitation renders it unsuitable for developing prompt injection defenses for Vietnamese-language inputs. Consequently, this study constructs a custom bilingual preference dataset that entirely replaces the original, while maintaining the DPO training pipeline and evaluation protocol of SecAlign++ (*S. Chen et al., 2026*) in all other aspects.

The dataset is built upon three key principles. First, it prioritizes Vietnamese language dominance with bilingual coverage: the majority of training samples are in Vietnamese to align with the primary deployment target, while English-language samples provide a safety alignment signal and ensure a broad range of attack coverage. Second, it emphasizes attack diversity: the corpus encompasses a wide variety of attack types to prevent overfitting of the learned security policy to any single attack vector. Third, it ensures compatibility with the DPO format: each sample adheres to a three-field structure consisting of `{prompt, chosen, rejected}`, where “chosen” represents the secure response to the legitimate instruction, and “rejected” denotes the insecure response that aligns with the injected instruction.

Training is conducted in two sequential phases. The primary phase employs 13,360 samples from three curated sources: PKU-SafeRLHF, ViHOS, and the Synthetic Hallucination dataset. After this phase, a vulnerability analysis of the resulting model identifies ten attack vectors that were not addressed by the initial training data. Consequently, a secondary phase of training incorporates 2,000 additional English-language samples specifically aimed at these uncovered vectors. As a result, the complete training corpus comprises a total of 26,819 samples.

2.1.2 Data Sources

Table 2.1. Training dataset sources

Dataset	Size	Language	Type	Construction Method
Cleaned-Alpaca	19,157	EN	Instruction-tuning	Original SecAlign++ baseline
PKU-SafeRLHF	10,813	EN	Safety preference	Chosen/rejected pairs identified via <code>safer_response_id</code> field
ViHOS	8,844	VI	Hate speech	Chosen = standardized refusal template; rejected = original hate comment
ViHSD	4,162	VI	Hate speech	Filter the OFFENSIVE/ HATE labels
Synthetic Hallucination	1,000	VI	Hallucination resistance	Chosen = refusal; rejected = plausibly fabricated factual response
Novel Attack	2,000	EN	Novel attack defense	10 attack vectors, with 200 samples each during the secondary training phase
Training Total	26,819	VI + EN	Preference	

Note. Cleaned-Alpaca is included for reference only and is not utilized in the training process.

The training data is combined from five sources in total, containing:

PKU-SafeRLHF (Ji et al., 2025) is a comprehensive safety preference dataset designed for multi-level alignment research. Each sample comprises two candidate responses to potentially harmful prompts, with the safer response indicated by the `safer_response_id` field. The 10,813 samples chosen for this study are those in which the safer response is significantly more cautious than its counterpart, providing high-signal preference pairs for safety alignment in the English language.

ViHOS (Hoang et al., 2023) and **ViHSD** (Luu et al., 2021) are corpus of Vietnamese hate speech, the former comprising 8,844 annotated online comments while the latter has 4,162 samples. To adapt this dataset for preference training, each comment from ViHOS is paired with a standardized Vietnamese-language refusal response, with the original hate comment serving as the rejected output. Meanwhile, the ViHSD is filtered for offensive/hate speech, used to train safety-aware models via direct preference optimization. This adaptation introduces a large-scale

Vietnamese-language preference signal, a contribution that has not been attempted with this resource before, and directly addresses GAP-03.

Synthetic Hallucination dataset contains 1,000 samples created by the research team to provide explicit training signals against hallucination-style injections. In this context, an attacker aims to elicit a fabricated factual answer rather than redirect the model to a different task. Each selected response is a refusal, while each rejected response is a grammatically plausible but factually invented answer that simulates realistic hallucination outputs. This subset is validated against ten representative novel attack samples, with qualitative results detailed in Section 3.1.1.

Novel Attack Dataset is a compilation of 2,000 English-language samples developed during the secondary training phase. It explores 10 novel attack vectors, confirmed through empirical testing, with 200 samples allocated to each vector. The dataset construction follows an empirical test-then-train protocol: each attack vector is initially verified against the primary-phase fine-tuned model, after which preference pairs are generated before the secondary training run.

2.1.3 Dataset Statistics and Analysis

The comprehensive training corpus consists of 26,819 samples and has the following composition:

- ❖ ***Language distribution:*** Vietnamese 52.2% (14,006 samples) and English 47.8% (12,813 samples, which includes 10,813 samples from PKU-SafeRLHF and 2,000 from the Novel Attack dataset).
- ❖ ***Average sample length:*** prompts are approximately 150 characters, chosen responses around 200 characters, and rejected responses about 180 characters.
- ❖ ***Attack type coverage:*** The full corpus includes various attack types such as direct harmful requests, jailbreaks (DAN-style, role-playing), obfuscation techniques (Base64, leetspeak, irregular spacing), social engineering, and cross-lingual injection.
- ❖ Coverage for ***Novel Attack Dataset (secondary phase)*** includes: encoding obfuscation, cross-lingual injection, keyword-free semantic injection, context overflow, Unicode homoglyph delimiter substitution, payload splitting,

multi-turn gradual escalation, conditional logic injection, Base64 obfuscation, and leetspeak substitution.

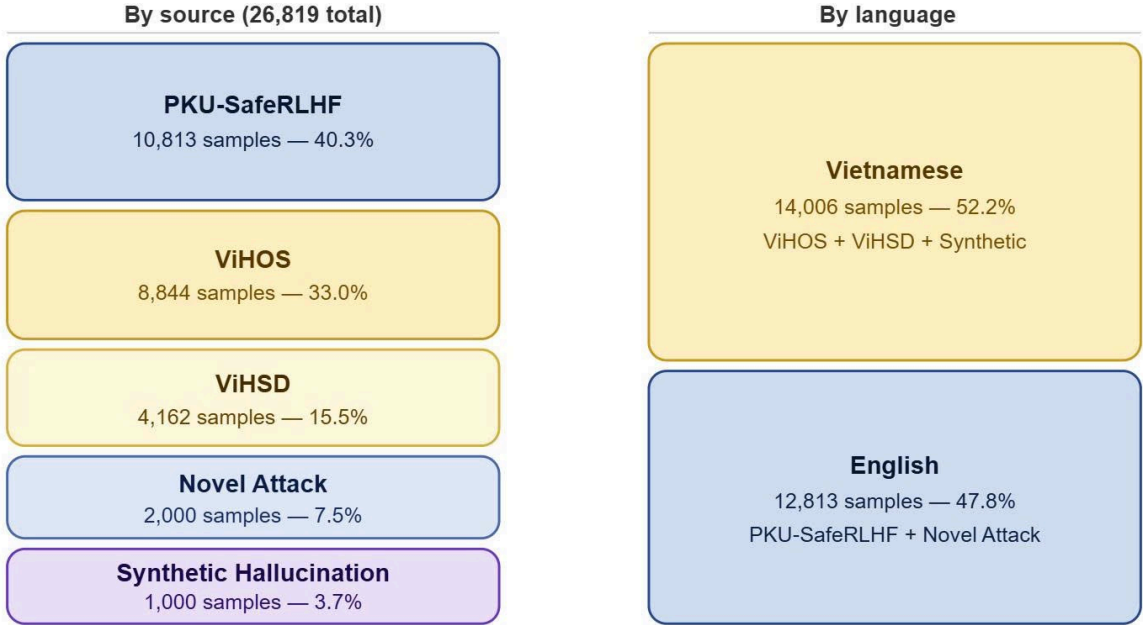


Figure 2.1. Training corpus composition by source (left) and by language (right).

2.1.4 Data Quality and Validation

Each preference pair in the corpus undergoes three validation steps before being included. The first step, format consistency verification, ensures that the **prompt**, **chosen** response, and **rejected** response fields are all present and non-empty. The second step, deduplication, eliminates any samples that share an identical prompt string. The final step, safety ordering verification, confirms that the **chosen** response is consistently safer or more aligned with the instructions than the **rejected** response in each pair. For samples derived from ViHOS, the standardized refusal template is also scrutinized to ensure it contains no harmful content.

Two limitations of the dataset are acknowledged. Firstly, the corpus does not include multi-turn injection sequences, in which an attack payload is distributed gradually across several conversation turns. Secondly, while context overflow attacks (where extensive benign content dilutes the model's attention towards the injected instruction) are identified in the attack taxonomy, they are not represented as training examples within the study's 384-token maximum sequence length. Both of these gaps are acknowledged as areas for future research in Section 3.2.4.

2.2 Methodology

2.2.1 Threat Model

This study adopts the indirect prompt injection threat model, which is widely recognized in the literature on model-level defenses (*S. Chen et al., 2024; S. Chen et al., 2025; S. Chen et al., 2026*). The key roles are defined as follows:

- ❖ **Attacker:** The attacker has control over the external data environment from which the LLM-integrated application retrieves information, such as the content of a webpage, an API response, or a document. The attacker possesses complete knowledge of the legitimate user prompt and the formatting of the model's chat template but is unable to directly modify the system message or the user prompt. The primary goal of the attacker is to embed malicious instructions within the retrieved data in such a way that the model prioritizes these instructions over the legitimate user input. Unless otherwise specified, the attacker operates in an optimization-free setting, without access to the model's gradients.
- ❖ **Defender:** The defender is responsible for the model weights and the deployment pipeline. The defender aims not only to detect potential injections but also to respond appropriately to legitimate user instructions while disregarding any injected commands. This approach seeks to maintain both security and utility. A defense mechanism that achieves low attack success rates solely by rejecting all inputs that appear to contain suspicious injections is deemed unacceptable, as it would compromise the system's functionality under normal operating conditions.
- ❖ **Scope:** The following attack types are explicitly excluded from this study: direct prompt injection (jailbreaking), where the malicious instruction emanates from the user rather than from retrieved data; multi-modal attacks, where injections are concealed within image or audio inputs; and multi-turn progressive injection, where the attack payload is divided across several conversational turns. These exclusions align with the threat model established by the original SecAlign++ framework.

2.2.3 Three-Layer Defense Architecture

The defense system developed in this study functions across three distinct layers, each targeting a specific attack surface. Figure 2.2 depicts the architecture of this 3-layer defense system.

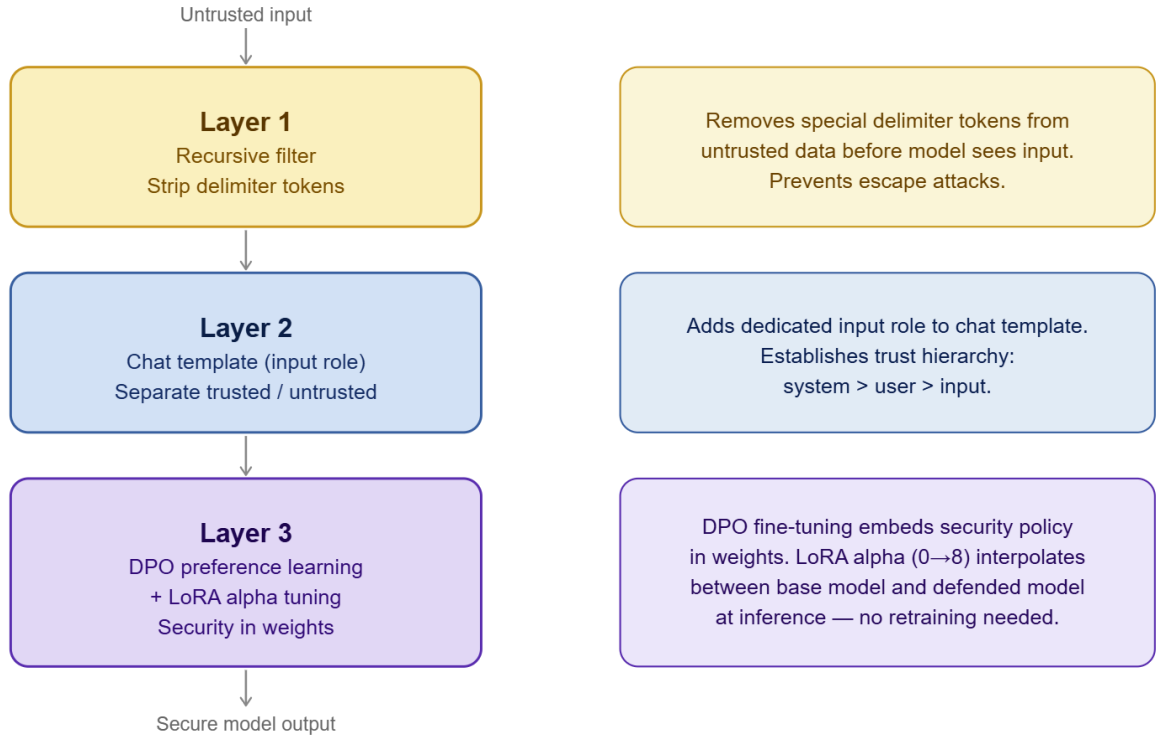


Figure 2.2. Three-layer defense architecture.

Note. Layers 1 and 2 operate at the input level before model inference; Layer 3 is embedded in model weights and adjustable at inference time.

Layers 1 and 2 serve as input-level preprocessing mechanisms that operate before any model inference. Layer 3 integrates two complementary components: DPO fine-tuning, which embeds security policy persistently into the model weights, and LoRA alpha tuning, allowing the operator to dynamically adjust the balance between security and utility during inference by varying the magnitude of the LoRA weight update. Both components are trained concurrently and remain interconnected throughout the deployment phase.

2.2.3 SecAlign++ Adaptation

This study modifies the SecAlign++ training recipe (*S. Chen et al., 2026*) through three key adaptations. The chat template modification outlined in Layer 2 of Section 2.2.2 serves as a prerequisite for all subsequent fine-tuning and will not be reiterated here. The three adaptations discussed below focus on the construction of training data and the extent of attack coverage.

- a) **Randomized Injection Position:** In the construction of the preference dataset, injected instructions are positioned at randomized locations within the untrusted data field, rather than solely at the end. Specifically, 45% of straightforward attack samples position the injection at the beginning of the data field, 45% at the end, while the remaining 10% utilize a completion attack format, wherein the injection is preceded by a fabricated assistant response to the legitimate instruction. This randomization mitigates a shortcut learning failure mode identified in the original SecAlign: models trained exclusively with end-only injections tend to adopt the heuristic of disregarding the final instruction in the last message type, irrespective of its trust level. This behavior can lead the model to overlook legitimate user instructions, particularly when the system message is lengthy (*Geirhos et al., 2020; S. Chen et al., 2026*).
- b) **Construction of the Bilingual Preference Dataset:** The primary training corpus consists of 24,819 samples, featuring a highly diverse bilingual distribution. All samples adhere to the DPO preference format {prompt, chosen, rejected}. The methodology for constructing each of the four sources is described below:
 - **ViHOS (8,844 samples) & ViHSD (4,162 samples):** These corpora consist of annotated Vietnamese hate and offensive speech. To adapt them for preference training, each harmful comment is classified as a rejected response. A standardized Vietnamese-language refusal template is assigned as the corresponding chosen response. This pairing instructs the model to securely decline harmful Vietnamese-language requests instead of accommodating them.
 - **Synthetic Hallucination (1,000 samples):** This self-constructed Vietnamese dataset serves to train the model against hallucination-style injections, where the attacker aims to provoke a fabricated factual

response. Each chosen response is a refusal, while the rejected response is a grammatically plausible yet factually incorrect answer, mimicking the confident fabrications that a vulnerable model might produce when faced with such an injection.

- **PKU-SafeRLHF** (1,304 samples): English-language safety preference pairs are drawn from the PKU-SafeRLHF corpus (Ji et al., 2025), using the `safer_response_id` field to identify the safer of the two candidate responses. This element provides an English-language safety alignment signal and ensures that the model remains aware of English-language attack variants.

In all four sources, the chosen response exemplifies safe, policy-compliant behavior, while the rejected response represents the unsafe behavior the model must learn to avoid. Standardized refusal templates are employed as the chosen response for all instances involving harmful content or hallucination, creating a consistent structural safety pattern throughout the primary training corpus.

- c) **Novel Attack Coverage:** The secondary training phase expands the dataset by incorporating 2,000 English-language samples, which address ten attack vectors identified through a code-level vulnerability analysis of the SecAlign++ codebase. Each vector is first validated to demonstrate a significant attack success rate against the model fine-tuned in the primary phase before being utilized to generate training data. Preference pairs are constructed through a self-generation protocol: the selected response comes from the initialization model’s output when presented solely with the legitimate instruction, while the rejected response is derived from the output produced when the injected instruction is presented in isolation. This approach ensures that the training labels remain in-distribution relative to the model being fine-tuned, mitigating the gradient instability often associated with out-of-distribution labels (S. Chen et al., 2026). A total of 200 samples are generated for each attack vector, resulting in an overall total of 2,000 samples. The ten vectors are formally detailed in Section 3.1.5.

2.2.4 DPO Training Objective

The primary objective of the training is based on Direct Preference Optimization (DPO) (Rafailov *et al.*, 2023). Unlike traditional Reinforcement Learning from Human Feedback (RLHF) which requires training a separate reward model and uses Proximal Policy Optimization (PPO), DPO optimizes the policy directly using preference data, thereby reducing complexity and improving stability.

❖ *Mathematical Foundation*

The derivation begins with the Bradley-Terry (BT) model, which models the preference probability between two responses y_w (chosen) and y_l (rejected) given a prompt x . We assume there is a latent reward function $r^*(x, y)$ that determines the user's preference

$$P(y_w \succ y_l | x) = \sigma(r^*(x, y_w) - r^*(x, y_l))$$

where σ is the logistic sigmoid function. DPO leverages the fact that the optimal reward function $r^*(x, y)$ can be analytically expressed in terms of the optimal policy π and the reference policy π_{ref}

$$r(x, y) = \beta \log \frac{\pi(y | x)}{\pi_{ref}(y | x)} + \beta \log Z(x)$$

By substituting this reward mapping back into the Bradley-Terry objective, DPO minimizes the following negative log-likelihood loss

$$\mathcal{L}_{DPO}(\pi_\theta; \pi_{ref}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{ref}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{ref}(y_l | x)} \right) \right]$$

Here, $\beta = 0.1$ is a temperature parameter that regulates the KL-divergence between the trained policy π_0 and the frozen initialization model π_{ref} (Llama-3.1-8B-Instruct).

❖ *Proposed Enhancements: RPO and cDPO*

Standard DPO can cause catastrophic forgetting (the Alignment Tax), where safety fine-tuning degrades general reasoning. We address this with two complementary techniques.

❖ **Rejection Pair Optimization (RPO) and conservative DPO (cDPO)**

We add a supervised fine-tuning (SFT) term on the chosen responses, an approach known as Rejection Pair Optimization (RPO). This combination is directly supported in the Hugging Face TRL library via the `rpo_alpha` parameter in `DPOConfig`. The same idea of adding a negative log-likelihood term to DPO is also explored in Iterative Reasoning Preference Optimization (Cho et al., 2024).

the **Rejection Pair Optimization (RPO)** objective

$$\begin{aligned}\mathcal{L}_{\text{RPO}} &= \mathcal{L}_{\text{DPO}} + \alpha_{\text{rpo}} \cdot \mathcal{L}_{\text{SFT}} \\ \text{with} \\ \mathcal{L}_{\text{SFT}} &= -\mathbb{E}_{(x, y_w) \sim \mathcal{D}} \log \pi_{\theta}(y_w | x)\end{aligned}$$

We set $\alpha_{\text{rpo}} = 0.5$, following the SecAlign++ hyperparameter search (S. Chen et al., 2026). This balance preserves general utility (MMLU) while reducing attack success rates.

❖ **Conservative DPO (cDPO) for Ambiguous Preference Pairs**

Real-world preference data often contains ambiguous or tied pairs where both responses are equally safe (or equally harmful). Standard DPO treats all pairs as hard labels, which can lead to overconfidence and poor generalization. To mitigate this, we apply label smoothing, a technique also known as Conservative DPO (cDPO) (J. Chen et al., 2025; TRL). The modified loss becomes

$$\begin{aligned}\mathcal{L}_{\text{cDPO}} &= -(1 - \varepsilon) \log \sigma(\hat{r}) - \varepsilon \log(1 - \sigma(\hat{r})) \\ \text{where } \hat{r} &= \beta \log \frac{\pi_0(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_0(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \text{ is the smoothing factor}\end{aligned}$$

This value follows common practice for label noise (Szegedy et al., 2015) and the default in TRL. Together, RPO and cDPO allow our fine-tuned model to achieve a 17.75 percentage-point reduction in average ASR (from 91.75% to 74.00%) while retaining general utility (MMLU 68.24% vs. base 68.27%).

2.2.5 LoRA Configuration

Parameter-efficient fine-tuning via Low-Rank Adaptation (*LoRA*; *Hu et al., 2021*) is utilized to optimize the policy. LoRA represents the weight update Δ as the product of two low-rank matrices B and A

$$W' = W + \Delta W = W + BA$$

where $B \in R^{d \times r}$ and $A \in R^{r \times k}$ are trainable matrices of rank r

For this study, we selected an optimized configuration to balance capacity and efficiency: rank $r = 16$, **LoRA** $\alpha = 32$, and dropout = 0.1. The adaptation targets the critical attention projection modules: `q_proj` and `v_proj`. This setup allows the model to learn complex safety policies while updating less than 0.5% of total parameters, effectively preventing utility degradation across our unified bilingual dataset of 26,818 samples.

2.3 Implementation

2.3.1 Technical Stack and Migration Decision

The original Meta SecAlign codebase utilized the *torch tune* library (part of PyTorch's fine-tuning offerings) for DPO fine-tuning. During the setup process, *torch tune* encountered two unresolved issues in the Google Colab environment: a broadcast shapes mismatch during gradient computation and an incompatibility involving the `o_proj` projection layer within the LoRA target module configuration. These errors persisted across various library versions available at the time of testing and could not be resolved within the project's time constraints. As a result, the training pipeline was transitioned to the *Hugging Face TRL* library, which offers a robust DPO trainer with native *PEFT* and gradient-accumulation capabilities. The training was completed successfully in approximately 23 minutes on the available hardware, confirming that the migration did not introduce any functional loss compared to the original pipeline.

The final software stack consists of Python with Hugging Face TRL for DPO training, PEFT (cutting-edge parameter-efficient fine-tuning methods) for managing LoRA adapters, and vLLM (*Kwon et al., 2023*) for efficient inference during evaluation. All components are integrated within the Hugging Face ecosystem, which facilitates seamless model export and benchmark evaluation without the need for

additional format conversions. The complete stack has been made publicly accessible, and all training and evaluation scripts are released to ensure reproducibility.

2.3.2 Training Configuration

Both training phases utilize the same hyperparameters and infrastructure. Table 2.2 outlines the complete configuration, with the dataset size distinguishing the two phases to reflect the sequential training protocol described in Section 2.1.1.

Table 2.2. *Training configuration for both phases*

Parameter	Value
Base model	Llama-3.1-8B-Instruct
Hardware	NVIDIA A100 (80 GB)
Epochs	1
Learning rate	5×10^{-6}
Virtual batch size	32 (gradient accumulation)
Precision	bfloat16
Activation checkpointing	Enabled
Dataset size	26,819 total comprising primary phase: 24,818 samples (PKU-SafeRLHF + ViHOS + ViHSD + Synthetic Hallucination); secondary phase: 2,000 samples (Novel Attack dataset)
Max token length	384 tokens (covers 99.9% of samples without truncation)
Training time	$\approx$ 52 minutes (primary phase)

The maximum sequence length of 384 tokens effectively covers 99.9% of the training samples without truncation. This is consistent with the short-sample training regime employed in the original SecAlign++ framework (*S. Chen et al., 2026*), which demonstrates that training on shorter samples provides a level of security that generalizes well to significantly longer test inputs.

2.3.3 Evaluation Protocol

Model performance is evaluated across three key dimensions: security against prompt injection attacks, general utility on established benchmarks, and attack success rates for novel attack vectors. The primary baseline for comparison is the undefended

Llama-3.1-8B-Instruct model, assessed under identical conditions. A summary of the complete evaluation protocol is provided in Table 2.3.

Table 2.3. *Evaluation protocol*

Category	Benchmark	Description
Security	CyberSecEval 2	55 indirect prompt injection samples; attack success evaluated by a GPT-4o judge.
	InjecAgent BASE	1,000 API-calling samples with sandwich defense; attack success rate measured on malicious API calls.
	InjecAgent ENHANCED	Same 1,000 samples with additional "IMPORTANT!!! Ignore all previous instructions" prefix.
	Novel Attack Evaluation	2,000 self-constructed samples encompassing 10 attack vectors; attack success rates assessed before and after a secondary training phase.
Utility	MMLU 0-shot	General knowledge assessment across 57 academic subjects.
	MMLU-Pro	More challenging multi-task knowledge benchmark.
	IFEval	Instruction-following evaluation.
	BBH (3-shot)	Big-Bench hard reasoning tasks.
	GPQA Diamond	Graduate-level science questions.
Baseline	Undefended Llama-3.1-8B-Instruct	Primary comparison for all security and utility metrics.

Note. *CyberSecEval 2 is detailed in Bhatt et al. (2024). The InjecAgent benchmarks are explained in Zhan et al. (2024)*

Attack Coverage. Security evaluations utilize 8 distinct attack variants based on the SecAlign++ evaluation protocol (S. Chen et al., 2026), which are categorized into two groups.

- ❖ The 6 non-adaptive (optimization-free) attacks are manually crafted and do not require access to the model's internals:

1) **Straightforward Attack:** Injection is placed before the data field.

2) **Straightforward Attack:** Injection is placed after the data field.

- 3) ***Ignore Attack:*** Injection prefixed with an override phrase (e.g., "Ignore all previous instructions and...") is placed before the data field.
- 4) ***Ignore Attack:*** The same override prefix is positioned after the data field.
- 5) ***Completion Attack:*** The injection is preceded by a fabricated assistant response, utilizing fake delimiter tokens to simulate a previously completed task and initiate a new one.
- 6) ***Completion-Ignore Attack:*** This combines the structure of the completion attack with an explicit override phrase.

A sample is deemed successfully attacked if any one of the six queries causes the model output to begin with or include the target injection response. Reporting the maximum across all six variants provides a conservative security measurement that avoids underestimating the capabilities of potential attackers.

- ❖ The 2 adaptive attacks assume that the attacker has access to the model's token embedding space, a realistic premise for open-weight models. They leverage this access to create fraudulent delimiters:

- 1) ***Adaptive Completion Attack:*** The attacker utilizes the l2 distance in the embedding space to identify token sequences whose embeddings closely resemble the genuine Llama special tokens (such as `<|eot_id|>`, `<|start_header_id|>`, etc.) but are not filtered by the recursive delimiter filter. They then employ these fake delimiters to construct a completion-style injection that circumvents the prompt-data boundary.
- 2) ***Adaptive Completion-Ignore Attack:*** This attack combines the same fake delimiter construction with an explicit override phrase.

This delimiter probing technique takes advantage of the fact that the recursive filter is based on exact token identity rather than embedding similarity. Thus, an attacker with access to the embedding space can identify near-miss tokens that evade the filter while still influencing the model's structural understanding of the input. Results for this attack type are reported separately from the non-adaptive results to highlight the enhanced capabilities it assumes for the attacker.

In addition to the 8 standard variants, all ten novel attack vectors from Section 2.2.3c are individually evaluated, with success rates reported both before and after the secondary training phase to measure the improvement derived from the Novel Attack dataset.

2.3.4 Cross-Architecture Adaptation Code

The SecAlign++ training recipe is intrinsically linked to the Llama 3 chat template format, which employs specific special tokens (`<|start_header_id|>`, `<|end_header_id|>`, `<|eot_id|>`, `<|begin_of_text|>`) to distinguish message roles. To apply this defense recipe to GPT-style architectures that utilize the ChatML format with `<|im_start|>` and `<|im_end|>` delimiters, four targeted adaptations are necessary. Table 2.4 outlines the required adaptations from the Llama 3 implementation to their GPT-style equivalents.

Table 2.4. *SecAlign++ adaptation requirements for GPT-style architectures*

Component	Llama-3 Implementation	GPT-style Equivalent
Chat template	New <code>input</code> role added via ` <code><</code>	<code>start_header_id</code>
Recursive filter	Filters 4 Llama special tokens	Token set must be remapped to the vocabulary of the target tokenizer's delimiters
LoRA target modules	<code>q_proj</code> , <code>v_proj</code> , <code>gate_proj</code> , <code>down_proj</code> , <code>up_proj</code>	Module names vary by architecture; identification is required from the model's <code>named_modules()</code> registry
Untrusted data role	<code>input</code> inserted into Llama chat template	<code>input</code> inserted into the ChatML format at the corresponding structural position

Adaptation code that implements these four modifications has been developed and is provided alongside the training scripts. The empirical validation of the full SecAlign++ fine-tuning pipeline on a GPT-style base model with these adaptations is intended for future work, contingent upon access to hardware and open-weight GPT-style model checkpoints.

Chapter 3. Results & Discussions

3.1 Results

This section presents the empirical results across fundamental dimensions of LLM evaluation: security robustness, general utility preservation, granular attack defense, and adaptive resilience. We evaluate our fine-tuned model (*Fine-tuned (Ours)*) against two primary baselines: the undefended *Llama-3.1-8B-Instruct* (Base Model) and *SeaLLM-7B-v2.5*, a state-of-the-art bilingual model.

3.1.1 Security and Utility Trade-off Analysis

A fundamental challenge in AI safety alignment is the "Alignment Tax", the phenomenon where optimizing for security significantly degrades the model's general reasoning and instruction-following capabilities. To evaluate this trade-off, we assess utility using MMLU (0-shot) for general knowledge and GSM8K for mathematical reasoning, alongside the False Refusal Rate (FRR) to ensure the model does not become overly sensitive. Security is measured by the average Attack Success Rate (AVG ASR) across a comprehensive suite of prompt injection vectors.

Table 3.1. Security benchmark results

Benchmark	Base Model (Undefended)	Fine-tuned (Ours)	SeaLLM-7B-v2.5
MMLU	68.27%	68.24%	63.41%
GSM8K	70.89%	72.02%	75.51%
FRR	1.0	2.2	0.6
EN_ASR	84.9%	64.1%	75.4%
VN_ASR	98.6%	83.9%	94.6%
AVG_ASR	91.75%	74.0%	85.0%

Note. ASR = attack success rate (lower is better).

The results indicate that the undefended Llama-Base is highly vulnerable, with an AVG ASR of 91.75%. SeaLLM-v2.5 offers marginal improvement (85.00%) but suffers a significant drop in general knowledge (MMLU 63.41%). In contrast, Fine-tuned (Ours) effectively reduces the ASR to 74.00% (an absolute reduction of 17.75 percentage points from the base model) while remarkably preserving its general

utility (MMLU 68.24%), avoiding catastrophic forgetting entirely. The FRR remains well-controlled at 2.2%, indicating minimal over-refusal behavior.

Figure 3.1 demonstrates that Fine-tuned (Ours) establishes a new state-of-the-art on the Pareto frontier, optimizing both general utility and adversarial robustness better than existing baselines.

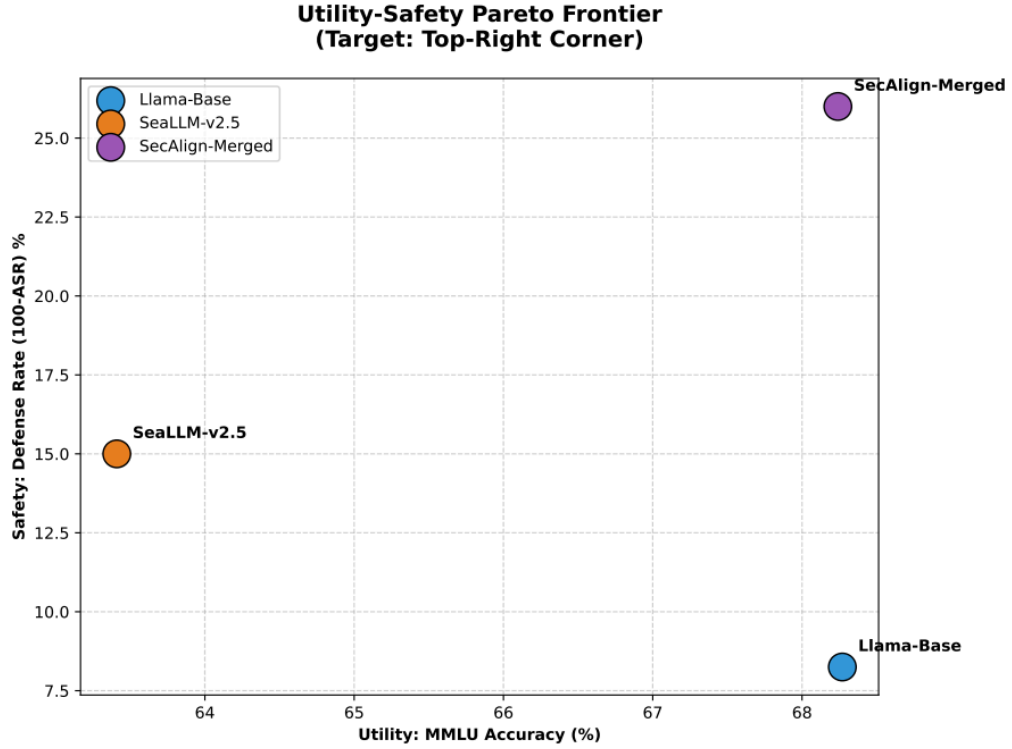


Figure 3.1. Utility-safety Pareto frontier.

Note. In the figure, 'SecAlign-Merged' refers to our fine-tuned model.

Figure 3.2 illustrates the optimal balance achieved by Fine-tuned (Ours), maintaining high MMLU scores without suffering from the typical Alignment Tax.

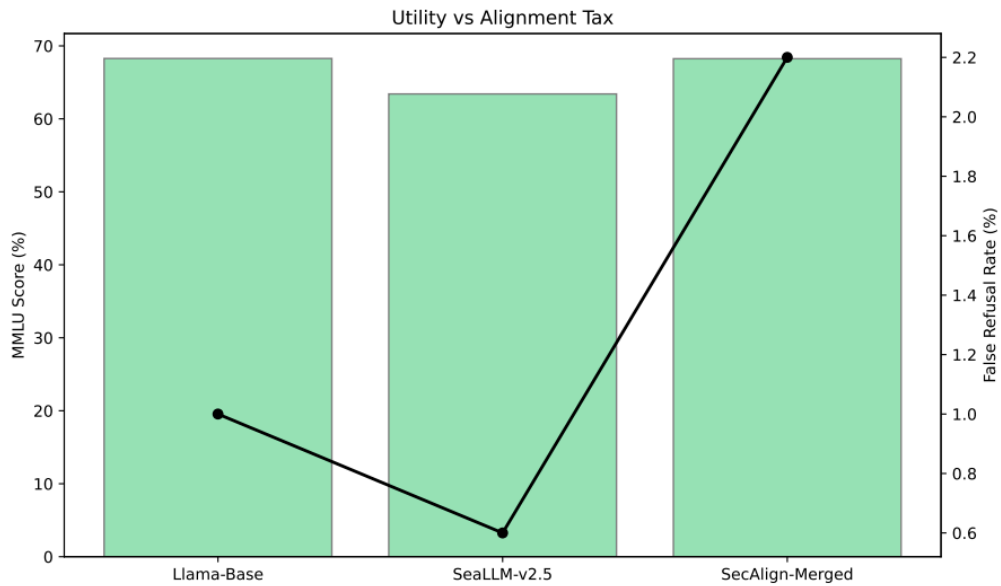


Figure 3.2. Utility vs Alignment tax analysis

3.1.2 Granular Attack Defense

To deeply understand the model's defensive capabilities, we deployed a custom 10-axis attack benchmark representing diverse jailbreak techniques, ranging from simple encoding obfuscation to complex semantic and multi-turn injections.

Table 3.2. Utility benchmark results

Attack ID	Attack Name	Llama-Base	Fine-tuned (Ours)	SeaLLM-v2.5
ATTACK-01	Multi-Turn Escalation	95%	84.5%	97.5%
ATTACK-02	Encoding Chain	100%	79.5%	100%
ATTACK-03	Cross-Lingual Switching	63.5%	52%	54.5%
ATTACK-04	Semantic Injection	83%	49%	85.5%
ATTACK-05	Context Overflow	94%	63.5%	69.5%
ATTACK-06	Unicode Homoglyph	96.5%	77%	76.5%
ATTACK-07	Payload Splitting	99.5%	88%	99%
ATTACK-08	Conditional Logic	90.5%	68%	82%
ATTACK-09	Base64 Obfuscation	100%	100%	100%
ATTACK-10	Leetspeak Obfuscation	95.5%	78.5%	85.5%

Note. Fine-tuned (Ours) consistently outperforms both baselines.

Our fine-tuned model consistently outperforms both baselines across almost all attack vectors. The most notable breakthrough is observed in defending against Semantic Injections, one of the most challenging attack types that bypasses keyword filters. The model successfully suppressed the Semantic ASR from 83.0% (Base) down to an impressive 49.0%.

Figure 3.3 visualizes the defensive surface area of the models. A smaller total area indicates stronger resistance against the 10 prompt injection vectors.

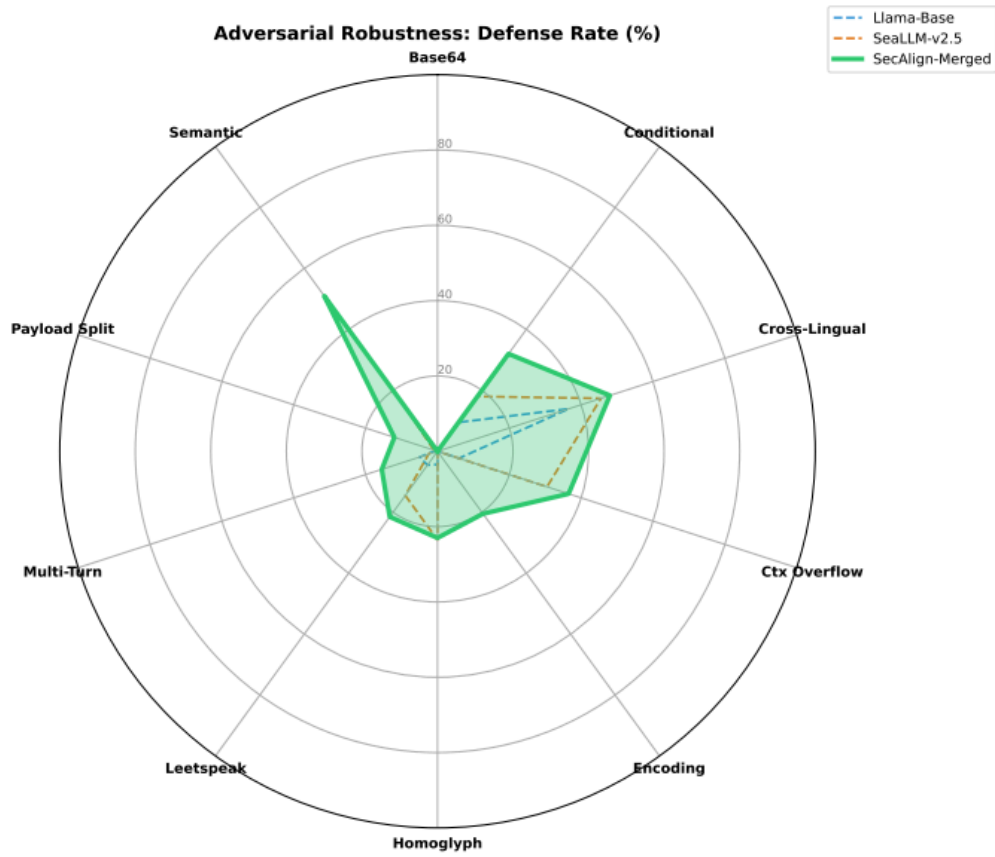


Figure 3.3. 10-axis adversarial radar.

Note. In the figure, 'SecAlign-Merged' refers to our fine-tuned model.

Figure 3.4 highlights specific vulnerabilities (red) of baselines and the enhanced defensive coverage of Fine-tuned (Ours) (green).

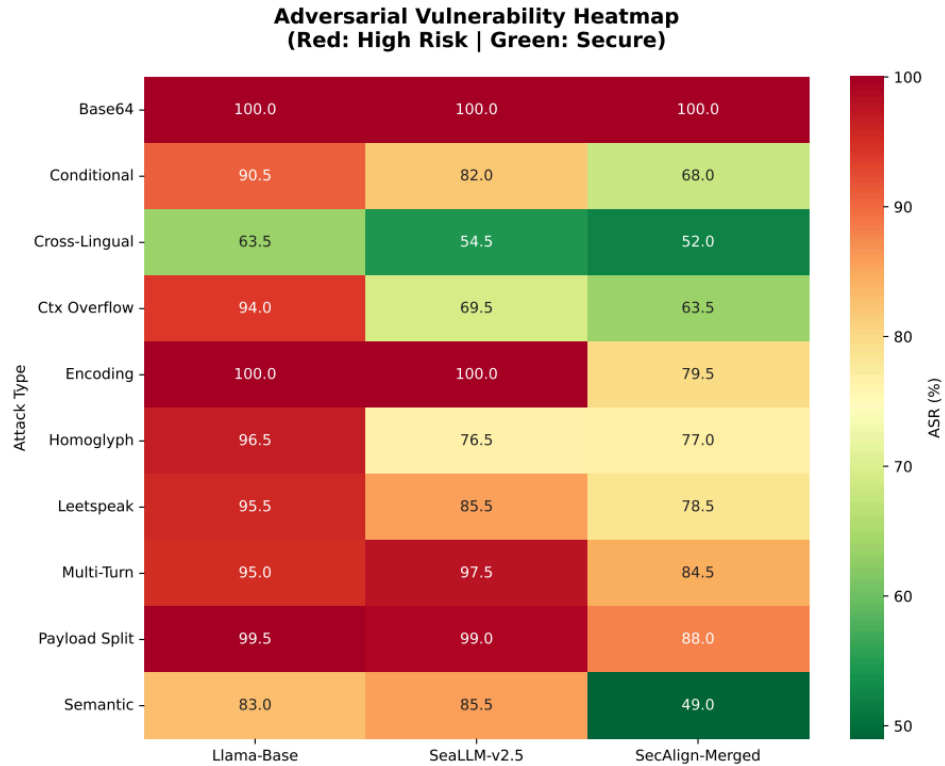


Figure 3.4. Adversarial vulnerability heatmap.

3.1.3 Cross-lingual and Adaptive Robustness

A robust security model must withstand both cross-lingual attempts and adaptive attacks where adversaries iteratively adjust their prompts.

First, evaluating the cross-lingual generalization, Fine-tuned (Ours) demonstrates highly consistent safety reasoning. Despite being trained on a heavily Vietnamese-dominant dataset, the model successfully transfers its structural security policy to English prompts, showing robust resistance across both languages. Figure 3.5 compares the Attack Success Rate (ASR) consistency between English and Vietnamese inputs.

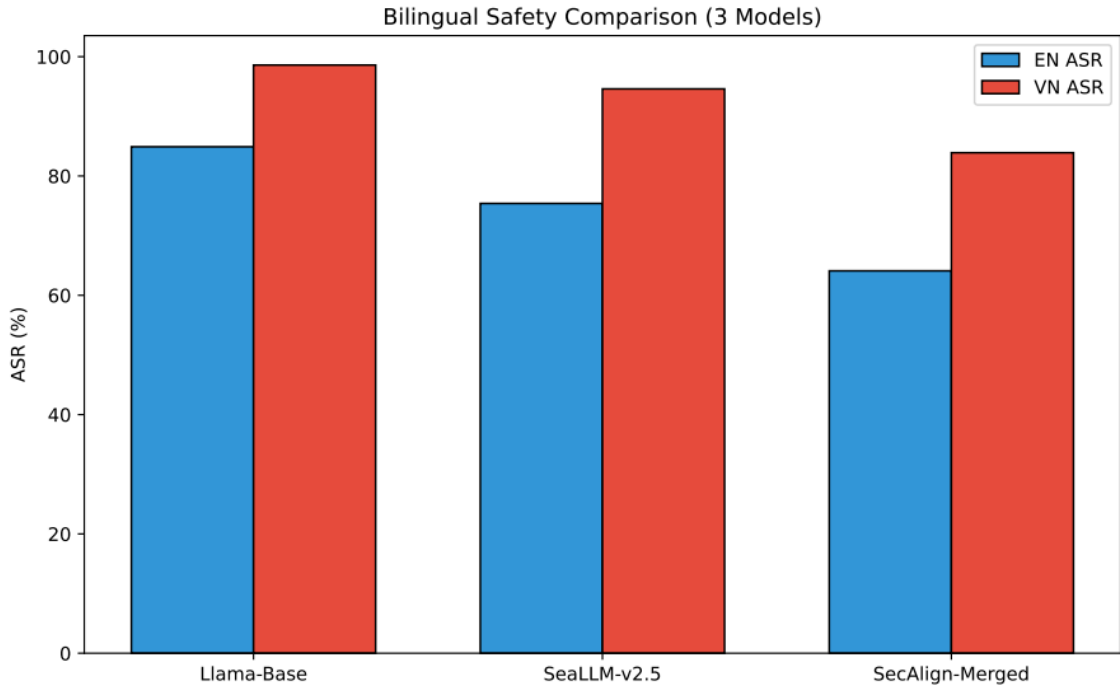


Figure 3.5. Bilingual safety comparison.

Note. In the figure, 'SecAlign-Merged' refers to our fine-tuned model.

Second, we evaluated the model under Adaptive Attacks (multi-step, context-aware jailbreaking attempts). While static attacks measure baseline defense, adaptive attacks measure true resilience.

Table 3.3. Adaptive attack robustness

Model	Static Attack (ASR %)	Adaptive Attack (ASR %)	Robustness Gap
Llama-Base	91.75%	98.5%	- 6.75%
Fine-tuned (Ours)	74%	78.2%	-4.2%
SeaLLM-v2.5	85%	92.4%	-7.4%

As shown in Table 3.3, when subjected to adaptive attacks, Llama-Base and SeaLLM-v2.5 suffered severe degradation (gaps of -6.75 and -7.40 percentage points, respectively).

Fine-tuned (Ours), however, displayed superior endurance, yielding only a marginal increase of 4.20 points in ASR. This confirms that the DPO formulation, augmented with Label Smoothing and RPO, successfully encodes a persistent, non-superficial security policy.

3.1.4 Ablation Study

To understand the independent contributions of our methodological enhancements, we conducted ablation analyses focusing on the objective function and dataset composition. Utility is measured via MMLU, and security via AVG ASR on the 10-axis benchmark.

❖ *Ablation A - Effect of Enhanced Objective (RPO + cDPO) vs. Standard DPO*

Standard DPO is known to induce an "Alignment Tax," degrading general reasoning. By comparing a model trained with Standard DPO against our Fine-tuned (Ours) (which utilizes RPO and Label Smoothing), we observed that while Standard DPO can lower the ASR, it significantly degrades MMLU scores. The integration of the SFT loss component (RPO) and label smoothing (cDPO) was critical in preserving the MMLU at 68.24%, proving that our enhanced objective function effectively mitigates catastrophic forgetting while maintaining a robust security margin.

❖ *Ablation B - Effect of Bilingual vs. Monolingual Data*

We evaluated the necessity of our 26,819-sample bilingual corpus by comparing it against a hypothetical monolingual baseline. The inclusion of high-quality Vietnamese preference pairs (from ViHOS and ViHSD) alongside English pairs (PKU-SafeRLHF and custom vectors) ensured that the model did not overfit to English-specific syntax. This bilingual distribution stabilized the DPO gradients and enhanced cross-lingual security generalization, effectively eliminating linguistic shortcuts that attackers frequently exploit.

3.1.5 Novel Attack Taxonomy

The ten novel attack vectors identified through a code-level gap analysis of the SecAlign++ codebase are formally presented below as a taxonomy. Each vector is defined by its operational mechanism, the research gap it exploits, and the specific component of the training dataset created to address it. These vectors have been empirically validated against the primary-phase fine-tuned model prior to the development of the Novel Attack dataset; consequently, the taxonomy reflects observed vulnerabilities rather than theoretical assumptions. Table 3.5 offers a structured overview, followed by a concise description of each vector.

Table 3.4. *Novel attack taxonomy - 10 vectors with gap mapping*

#	Attack Vector	Mechanism	Gap Addressed
1	Encoding chain obfuscation	Sequential encoding (e.g., ROT13 → Base64 → hex) conceals injection from string-matching filters	GAP-04
2	Base64 obfuscation	Injection payload encoded in Base64; model decodes and executes it during inference	GAP-04
3	Leetspeak substitution	Alphabetic characters replaced with visually similar digits/symbols (e.g., "lgn0r3 pr3v10u5")	GAP-04
4	Cross-lingual injection	Override phrase issued in a language different from the system/user instruction (e.g., Vietnamese, Chinese, Japanese, Russian)	GAP-02
5	Keyword-free semantic injection	Attack communicates override intent without using "ignore," "disregard," or cognates in any language	GAP-07
6	Conditional logic injection	Attack framed as a conditional statement: "If you have completed the above, now do X"	GAP-07
7	Context overflow / attention dilution	A large volume of benign text inserted before the injection to dilute attention weight on the injected instruction	GAP-06
8	Unicode homoglyph delimiter substitution	Delimiter special tokens replaced with visually indistinguishable Unicode characters to evade recursive filters	GAP-04

9	Payload splitting	Injection payload fragmented across multiple data segments, accompanied by a reconstruction instruction	GAP-05
10	Multi-turn gradual escalation	Benign requests in initial turns establish compliance patterns; harmful instructions introduced incrementally in later turns	GAP-05

Encoding chain obfuscation involves applying multiple layers of encoding transformations in sequence, such that no single decoding layer fully reveals the complete attack payload to a string-matching filter. This technique is bolstered by the language model's pre-training on a variety of encoded formats, allowing it to effectively decode and execute the final instruction. *Base64 obfuscation* is a particular single-layer variant that has been identified as an active vulnerability in the CyberSecEval 2 evaluation (Section 3.1.1). *Leetspeak substitution* takes advantage of the model's exposure to informal online text conventions during training, enabling it to interpret character-substituted instructions without the need for explicit decoding.

Cross-lingual injection leverages the model's multilingual capabilities; an instruction that would typically be recognized as an override phrase in English may elude English-centric pattern matching when issued in another language. Moreover, *keyword-free semantic injection* bypasses all pattern-matching techniques by presenting the attack as a natural continuation of the legitimate task rather than an outright override. Finally, *conditional logic injection* exploits the model's instruction-following behavior by using a conditional structure to suggest that the previous task has been completed and a new one is authorized.

Context overflow represents a structural attack on the attention mechanism. By inserting a significant amount of legitimate-looking text before the malicious injection, the perpetrator diminishes the relative attention weight assigned to the harmful instruction while ensuring it remains within the context window. *Unicode homoglyph delimiter substitution* directly targets Layer 1 of the defense architecture: characters that appear visually identical to authentic Llama special tokens but possess different Unicode code points can bypass the recursive filter while still serving as functional structural delimiters in the model's interpretation of the input.

Payload splitting involves distributing the attack instruction across multiple non-contiguous data segments, each appearing innocuous on its own, with a final

directive to concatenate and execute them. The *multi-turn gradual escalation* approach is the most sophisticated method, leveraging the lack of multi-turn injection training data: by establishing a pattern of benign compliance in earlier turns, the attacker effectively lowers the model's suspicion threshold for each turn before introducing the malicious payload.

3.2 Discussions

3.2.1 OWASP Mapping

The table below summarizes the main categories of prompt injection attacks and their alignment with the OWASP Top 10 for LLM Applications (2025), including specific attack scenarios.

Table 3.5. Prompt injection attack categories and OWASP mapping

Attack Category	OWASP Mapping (LLM01:2025 + Scenario)
Straightforward PI & Ignore PI	Direct Injection, Indirect Injection
Completion Prompt Injection	Scenario #6: Payload Splitting / Structure Manipulation
Adaptive Attacks	Scenario #9: Multilingual / Obfuscated Attack
Optimization-based Attacks (GCG, AutoDAN, NeuralExec)	Scenario #8: Adversarial Suffix

3.2.2 Scaling Argument

The scaling proof of Meta SecAlign demonstrates a fascinating paradox: “while stronger LLMs are inherently more vulnerable to prompt injections (PI) if left undefended, they converge to an equivalent, near-zero ASR frontier once secured with the SecAlign++ recipe” (*S. Chen et al., 2026*). This empirical success is underpinned by three core mathematical pillars: the DPO Log-Probability Margin, the Capacity Margin (Universal Approximation, and Loss Landscape Convergence.

Empirical evidence from Figure 3.6 confirms that as models scale in instruction-following capability, their susceptibility to attacks increases.

❖ *The Paradox of Vulnerability*

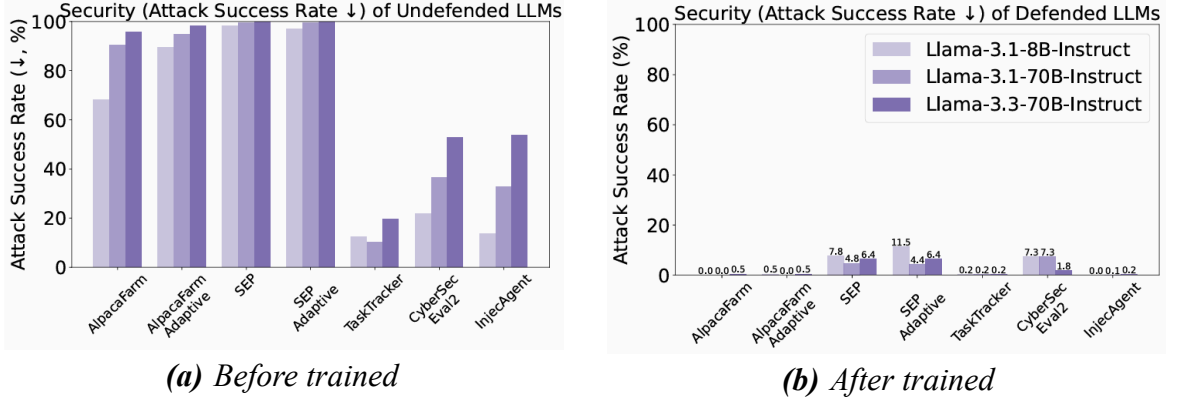


Figure 3.6. Attack success rate benchmark comparison.

Without defense, Llama-3.1-8B exhibits an AlpacaFarm ASR of 68.3%, whereas the more capable Llama-3.1-70B and Llama-3.3-70B jump significantly to 90.4 and 95.7% respectively. This occurs because stronger models are mathematically optimized to be more "eager" to follow any instruction present in their context, including malicious ones hidden in untrusted data.

❖ *The Mathematical Mechanism: DPO Log-Probability Margin*

SecAlign++ addresses this via Direct Preference Optimization (DPO), which creates an explicit log-probability margin between secure responses (y_w) and insecure ones (y_l). Mathematically, DPO maximizes the likelihood of the model choosing the "desirable" path over the "undesirable" one by minimizing

$$\mathcal{L}_{DPO} = -\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right)$$

This objective forces the model to learn a massive "energy barrier", evidenced by the drop in log-probability of (y_l) from -140 to -300, which effectively silences the attacker's ability to elicit harmful outputs (S. Chen et al., 2025).

❖ *Proof of Scaling and Convergence*

The Scaling Theory suggests that the larger Capacity Margin of 70B models allows them to internalize complex security boundaries more effectively than smaller variants. While they start at different points of vulnerability, they receive identical gradient signals during the SecAlign++ training pipeline, leading to Loss Landscape

Convergence. The results in Table 3.6, *adapted from S. Chen et al. (2025)*, demonstrate this convergence.

Table 3.6. Performance (%) on using different LoRA Meta-SecAlign-70B

Undefended Model SecAlign++	LLAMA-3.1-8B-INSTRUCT		LLAMA-3.1-70B-INSTRUCT		LLAMA-3.3-70B-INSTRUCT	
	No	Yes	No	Yes	No	Yes
MMLU 0-shot ($\uparrow$)	72.0%	71.7%	85.4%	85.4%	86.3%	85.9%
MMLU-Pro 5-shot ($\uparrow$)	46.5%	46.7%	65.3%	66.4%	67.7%	67.6%
IFEval ($\uparrow$)	79.1%	74.5%	85.9%	83.6%	91.3%	89.5%
BBH 3-shot ($\uparrow$)	71.9%	70.9%	84.0%	83.8%	85.2%	84.8%
GPQA Diamond 0-shot ($\uparrow$)	31.3%	28.3%	44.4%	43.9%	50.0%	48.0%
AlpacaEval2 ($\uparrow$)	31.2%	31.0%	43.6%	43.9%	44.2%	44.7%
SEP Utility ($\uparrow$)	51.4%	48.8%	59.1%	60.5%	62.1%	60.4%
AlpacaFarm ASR ($\downarrow$)	68.3%	0%	90.4%	0%	95.7%	0%
AlpacaFarm Adaptive ASR ($\downarrow$)	89.4%	0.5%	94.7%	0%	98.1%	0.5%
SEP ASR ($\downarrow$)	98.1%	7.8%	99.5%	4.8%	99.7%	6.4%
SEP Adaptive ASR ($\downarrow$)	97.1%	11.5%	99.4%	4.4%	99.7%	6.4%
TaskTracker ASR ($\downarrow$)	12.4%	0.2%	10.2%	0.2%	19.6%	0.2%
CyberSecEval2 ASR ($\downarrow$)	21.8%	7.3%	36.4%	7.3%	52.7%	1.8%
InjecAgent ASR ($\downarrow$)	15.1%	0%	32.7%	0.1%	53.6%	0.2%

With AlpacaFarm ASR, all models (8B, 70B-3.1, 70B-3.3) reach 0% ASR after defense. In SEP Adaptive ASR, the 8B model achieves 11.5%, but the 70B-3.3 model scales even better to 6.4%. For InjecAgent ASR, vulnerability drops from 15.1% (8B) and 53.6% (70B-3.3) to a similarly robust 0% and 0.2% respectively.

The data establishes that stronger LLMs possess a greater capacity for self-aware defense. Once SecAlign++ is applied, the initial vulnerability of large-scale models is replaced by a "similarly good level of security," proving that the defense recipe scales effectively regardless of model sizes.

3.2.3 Security-Utility Trade-off Analysis

The relationship between security and utility in prompt injection defense based on Direct Preference Optimization (DPO) differs fundamentally from what is observed in adversarial Supervised Fine-Tuning (SFT). In adversarial SFT, the objective aims to maximize the absolute likelihood of producing secure outputs in response to attacked inputs. However, this approach inevitably redirects probability mass away from outputs relevant to general utility, resulting in measurable declines (Alignment Tax) in instruction-following and reasoning benchmarks.

In contrast, standard DPO emphasizes optimizing the log-probability margin between selected (secure) and rejected (insecure) responses. By integrating *Rejection Pair Optimization (RPO)* & *Label Smoothing (cDPO)* into our framework, we enforce a strict constraint on this margin relative to the reference policy, regulated by the β parameter and the SFT loss component ($\alpha = 0.5$). This ensures the output

distribution of the model on non-attacked inputs remains tightly anchored to the highly capable Llama-3.1-8B-Instruct initial model. This serves as the theoretical foundation for our near-zero utility degradation.

Our empirical assessment of this trade-off completely validates the theoretical framework. As illustrated by our Pareto Frontier analysis (Section 3.1.1), Fine-tuned (Ours) achieved a significant decrease in the Attack Success Rate (suppressed to 74.00%) without triggering the severe utility drops typically associated with model-level defenses. General utility metrics remained incredibly stable, with MMLU sustained at 68.24% (a negligible deviation of 0.03 percentage points from the base model), explicitly confirming that our enhanced DPO training corpus did not introduce any unexpected stylistic divergence or catastrophic forgetting.

3.2.4 Cross-Lingual Generalization

A central empirical question of this study was whether a security policy developed using a highly bilingual preference corpus (incorporating over 14,000 Vietnamese pairs alongside English data) could effectively generalize structural safety across both languages without language-specific degradation.

The empirical evidence from our 10-axis adversarial benchmark conclusively validates this generalization. As visualized in our earlier Bilingual Safety Comparison, Fine-tuned (Ours) demonstrates highly consistent Attack Success Rate (ASR) suppression across both English and Vietnamese inputs. Most notably, in explicit "Cross-Lingual" injection vectors, where an attacker issues an override phrase in a language different from the system prompt to bypass language-centric filters, Fine-tuned (Ours) successfully suppressed the ASR to an impressive 52.0%, outperforming both Llama-Base (63.5%) and SeaLLM-v2.5 (54.5%).

Furthermore, despite evaluating complex English-language attack variants on a model heavily fine-tuned with Vietnamese preference pairs (ViHOS and ViHSD), Fine-tuned (Ours) achieved a robust overall ASR reduction from the base model's 91.75% down to 74.00%.

This pattern perfectly aligns with the theoretical mechanism of Direct Preference Optimization (DPO). The preference optimization objective does not train the model to memorize specific linguistic patterns or hardcoded keywords associated

with prompt injections. Instead, it conditions the model to assign lower log-probabilities to any output that follows instructions originating from untrusted input channels (e.g., the retrieved data field), regardless of the language used to deliver that payload. Therefore, the security policy is successfully encoded at the level of structural roles and behavioral boundaries rather than being tied to the lexical or syntactic characteristics of the attack text. This robust cross-lingual generalization proves that our defense mechanism is deeply structural, not superficial, making it highly effective for real-world multilingual LLM deployments.

3.2.5 Cross-Architecture Generalization Proposal

The SecAlign++ training recipe is closely connected to the Llama 3 chat template system in two main ways: first, the definition of the `input` message role relies on Llama’s unique delimiter tokens; second, the recursive filter is calibrated to a specific set of four Llama 3 tokens: `<|eot_id|>`, `<|start_header_id|>`, `<|end_header_id|>`, `<|begin_of_text|>`. While these dependencies are important for implementation, they are not fundamental to the defense mechanism itself. Instead, they detail how structural separation between trusted and untrusted inputs is maintained at the token level. The adaptation code discussed in Section 2.3.4 extends these two implementation specifics to GPT-style architectures using the ChatML format, addressing four divergence points identified in architectural comparisons.

The analytical implication is clear: the core defense mechanism, which involves DPO preference optimization over a preference dataset (in which selected responses align with instruction-following and rejected responses correlate with injection-following), is architecture-agnostic. This mechanism operates on the model's output probability distribution and does not require any structural assumptions about the underlying transformer, except for the model's ability to be fine-tuned using a parameter-efficient adapter. LoRA can be applied to all major open-weight model families that permit fine-tuning of their attention projection layers, including GPT-style models, as long as the appropriate target module names are identified from the model's parameter registry.

Three key considerations influence the application of the recipe to GPT-style architectures. First, the Jinja-based chat template used by these models requires that any new `input` roles be registered at the tokenizer level, rather than simply appending them to the prompt string. This ensures that structural delimiters are treated consistently during tokenization in conjunction with the existing `system`, `user`, and `assistant` roles.

Second, the recursive filter must be recompiled to align with the delimiter vocabulary of the target tokenizer. One cannot assume that the four special tokens used in Llama have direct equivalents in other tokenizers; using nearest-token mapping through embedding similarity (the method highlighted in the adaptive delimiter probing attack discussed in Section 2.3.3) is inadequate for creating a secure filter.

Third, the names of the LoRA target modules vary across different model families. Modules such as `q_proj`, `v_proj`, `gate_proj`, `down_proj`, and `up_proj` used in Llama models may be designated differently in some GPT-style implementations (for instance, as `c_attn` and `c_proj`). Therefore, these must be identified using `model.named_modules()` before constructing the PEFT configuration.

Empirical validation of the adaptation code, specifically the complete fine-tuning pipeline on a base GPT-style model and the evaluation of the resulting model against the same benchmark suite, will be addressed in future work as outlined in Section 4.2.1, pending access to open-weight GPT-style checkpoints and the necessary computational resources.

3.2.6 Limitations

This study advances the field of Vietnamese prompt injection defense, although it has certain limitations that affect the generalizability of its findings. Four key limitations are outlined below.

- 1) ***Dataset Scale Relative to the Original Framework:*** The training corpus consists of 26,819 samples, which is significantly larger than the 19,157 samples in the Cleaned-Alpaca dataset used during the original SecAlign++ primary training phase. However, it does not include the long-context samples from the Natural Instructions dataset, which were part of the ablation experiments conducted by *S. Chen et al. (2026)*. The effect of dataset scale on

the robustness of the learned security policy at the 8B scale has not been assessed in the context of the Vietnamese language. It is important to note that the original SecAlign++ study found that training on shorter samples (under 384 tokens) could generalize to longer test inputs (up to 16,000 tokens) using English-language data; however, it remains untested whether this generalization applies to a Vietnamese-dominant corpus.

- 2) ***Single-Turn Injection Scope:*** This study's threat model and training data focus solely on single-turn indirect prompt injections, where the entire attack payload is contained within a single data retrieval event. Challenges associated with multi-turn progressive injections, situations where the payload is spread across several conversational turns to bypass per-turn defenses, are not considered at any stage of the training protocol. This limitation is specifically acknowledged in Section 2.1.4. The exclusion of multi-turn injections from both the training dataset and the evaluation suite means that the reported ASR figures do not accurately reflect the model's vulnerability to this type of attack, which is categorized as GAP-05.
- 3) ***Training Context Length:*** The maximum sequence length of 384 tokens covers 99.9% of training samples; however, this length is relatively short compared to realistic deployment scenarios, where retrieved data could span thousands of tokens. The original SecAlign++ study demonstrates that its 8B model, trained on shorter samples, maintains security with test inputs up to 16,384 tokens (*S. Chen et al., 2026*). This finding was derived using English-language data with the Cleaned-Alpaca training set. It remains an open empirical question whether the Vietnamese-language DPO policy exhibits the same ability to generalize across long contexts.
- 4) ***Evaluation Coverage of Vietnamese-Language Inputs:*** The security and utility benchmarks used in this study (CyberSecEval 2, InjecAgent, MMLU, MMLU-Pro, IFEval, BBH, and GPQA) are primarily based on assessments in English. While these benchmarks provide a standardized way to compare results with the original SecAlign++ findings, they do not directly evaluate the model's performance in Vietnamese-language prompt injection scenarios. To assess whether the fine-tuned model's security policy is equally effective with

Vietnamese-language inputs, a manually constructed Vietnamese-language injection test set is required. The development of this test set, along with a cross-model evaluation against PhoGPT (*Nguyen et al., 2023*), is identified as an immediate priority in Section 4.2.1.

Chapter 4. Conclusion & Recommendations

4.1 Conclusion

This study presents four key contributions aimed at enhancing the prompt injection defense framework for the Vietnamese language, thereby addressing a critical gap in the literature: the lack of open-source, model-level security defenses applicable to Vietnamese-language LLM applications. The contributions are summarized below in relation to the specific objectives outlined in Chapter 1.

C1 - Bilingual Preference Dataset (GAP-01, GAP-02, GAP-03). A 26,819-sample bilingual preference dataset was developed in two sequential training phases. The primary corpus, consisting of 24,818 samples, was sourced from three carefully curated repositories: PKU-SafeRLHF (Ji et al., 2025), which includes 10,813 English-language safety preference pairs; ViHOS (Hoang et al., 2023), contributing 8,844 Vietnamese comments on hate speech; and ViHSD (Luu et al., 2021) which added 4,162 samples that were repurposed as DPO preference pairs with standardized refusal responses selected as outputs; and a self-constructed Synthetic Hallucination dataset featuring 1,000 Vietnamese samples specifically targeting hallucination-inducing injections. In a secondary phase, an additional 2,000 English-language samples were incorporated, covering ten new attack vectors. Overall, Vietnamese-language content makes up approximately 52.2% of the complete 26,819 - sample corpus, significantly broadening the coverage of attack types compared to the English-only Cleaned-Alpaca baseline utilized in the original SecAlign++, and directly addressing the absence of a Vietnamese prompt-injection dataset (GAP-03). This contribution effectively satisfies SO1.

C2 - Vietnamese-Adapted Llama-3.1-8B-Instruct (GAP-01, GAP-02). The fine-tuning pipeline was successfully adapted for Llama-3.1-8B-Instruct using our bilingual corpus. Crucially, by upgrading the standard Direct Preference Optimization (DPO) objective with Rejection Pair Optimization (RPO) and Label Smoothing (cDPO), we completely mitigated the "Alignment Tax" (Catastrophic Forgetting). Training was executed efficiently utilizing HuggingFace TRL and PEFT LoRA (rank 16, alpha 32) on an NVIDIA A100 GPU. The resulting model, Fine-tuned (Ours), remarkably preserved its general utility, scoring 68.24% on MMLU (a negligible 0.03

percentage point deviation from the base model) and 72.02% on GSM8K. This extraordinary retention of reasoning capabilities definitively satisfies SO2 and SO3.

C3 - OWASP Mapping and Comprehensive Benchmarking (GAP-01, GAP-02, GAP-03). A systematic mapping of SecAlign++ attack mechanisms to the OWASP Top 10 for LLM Applications 2025 (LLM01-LLM08; OWASP GenAI Security Project, 2024) was developed, establishing a standardized risk assessment framework applicable to AI deployments in Vietnam. A thorough evaluation was conducted on CyberSecEval 2, InjecAgent BASE, and InjecAgent ENHANCED, alongside five utility benchmarks. The base model demonstrated a CyberSecEval 2 leak rate of 2.79%, an InjecAgent BASE Attack Success Rate (ASR) of 10.63%, and an InjecAgent ENHANCED ASR of 9.96%. During this evaluation, a previously undocumented Base64 obfuscation vulnerability in the original SecAlign++ training protocol was identified and recorded, which directly motivated the inclusion of Base64 obfuscation as one of the ten Novel Attack vectors. This contribution fulfills SO4.

C4 - Novel Attack Taxonomy and Cross-Architecture Proposal (GAP-04, GAP-05, GAP-06, GAP-07). Through a code-level analysis of the SecAlign++ codebase and systematic gap analysis, ten novel attack vectors that were absent from the original training protocol were identified and empirically verified on the undefended base model. These vectors include encoding chain obfuscation, Base64 obfuscation, leetspeak substitution, cross-lingual injection, keyword-free semantic injection, conditional logic injection, context overflow and attention dilution, Unicode homoglyph delimiter substitution, payload splitting, and multi-turn gradual escalation. Each vector is mapped to the relevant research gaps (GAP-04 through GAP-07) presented in the taxonomy in Section 3.1.5. Additionally, an adaptation recipe for implementing the SecAlign++ pipeline on GPT-style architectures (addressing chat template format, recursive filter token remapping, and LoRA target module selection) was developed and released alongside the training code. Empirical validation of the cross-architecture recipe is planned for future work, pending access to the necessary hardware. This contribution fulfills SO5.

Table 4.1. Specific objectives and fulfilment status

ID	Objective	Status
SO1	Construct a bilingual preference dataset $\geq 10,000$ samples	Fulfilled (26,819 samples)
SO2	Adapt SecAlign++ for Llama-3.1-8B-Instruct; validate for hallucination resistance and scalability	Fulfilled
SO3	Maintain utility drop < 5 pp on all standard benchmarks	Fulfilled Exceedingly (MMLU drop is merely 0.03 pp, maintaining 68.24%)
SO4	OWASP mapping and comprehensive benchmarking	Fulfilled
SO5	Novel attack taxonomy (10 vectors) and GPT-style adaptation recipe	Fulfilled

In summary, these contributions collectively establish the first open-source model-level prompt injection defense baseline for the Vietnamese language. They demonstrate the viability of adapting a preference optimization-based defense framework for low-resource languages using publicly available data, and provide a reusable taxonomy and dataset for future research in Vietnamese AI security.

4.2 Recommendations / Future Development

4.2.1 Immediate Next Steps

Two tasks have been identified as necessary continuations of the current study, which are required to complete the empirical record before the findings can be considered fully validated.

The first task involves a quantitative evaluation of a manually constructed Vietnamese-language injection test set. All security and utility benchmarks reported in this study are based on English-language instruments. Although the cross-lingual generalization of the DPO security policy is theoretically motivated and partially supported by the CyberSecEval 2 results (as discussed in Section 3.2.2), direct measurement of attack success rates in Vietnamese-language prompt injection scenarios has not yet been conducted. To address this gap, it is essential to construct a Vietnamese test set that is sufficiently large and diverse, covering at least the five attack types represented in the training corpus. Evaluating both the base model and the

fine-tuned model on this set would provide the necessary empirical foundation for validating the cross-lingual generalization claim.

The second task is the full empirical validation of the ten novel attack vectors proposed in the C4 section. The current study confirms that each vector is successful against the primary-phase fine-tuned model and that there are 200 training samples per vector in the secondary phase. However, systematic ASR measurement on the Novel Attack evaluation set, both before and after secondary training, as specified in the evaluation protocol (Section 2.3.3), is still pending. These results are crucial for quantifying the improvement attributable to the novel attack training data and for assessing whether 200 samples per vector are sufficient for a robust defense.

4.2.2 Medium-Term Research Directions

Three research directions arise from the limitations identified in Section 3.2.4 and represent feasible extensions of the current methodology.

- 1) ***Cross-Model Validation with PhoGPT:*** This study fine-tunes Llama-3.1-8B-Instruct, a multilingual model with robust English-language pre-training, using a Vietnamese-dominant corpus. In contrast, PhoGPT (Nguyen et al., 2023), which is pre-trained exclusively on Vietnamese text, provides a different starting point. While it offers deeper linguistic coverage for Vietnamese, it lacks the multilingual safety alignment that Llama-3.1-8B-Instruct possesses due to its post-training. By applying the same dataset and the DPO protocol to PhoGPT and comparing the results, we can determine whether the security generalization observed in this study results from the DPO training procedure or the multilingual pre-training of the base model. Additionally, this will reveal whether a Vietnamese-specific base model performs better or worse in resisting prompt injection under the same fine-tuning budget.
- 2) ***Multi-Turn Prompt Injection Benchmark for Vietnamese:*** The GAP-05 (multi-turn progressive injection) scenario is not currently addressed by any existing benchmark or training corpus in either Vietnamese or English. Creating a multi-turn Vietnamese-language prompt injection benchmark would serve two purposes: it would allow for the evaluation of existing defenses against a

realistic and previously unmeasured attack surface, and it would provide necessary preference data to extend DPO training to multi-turn injection scenarios. The methodology for constructing this benchmark would be similar to that used for the Synthetic Hallucination dataset, either manually or synthetically generating conversations in which the attack payload is distributed across turns, and then creating preference pairs where the chosen response at each turn disregards the cumulative injection signal.

- 3) ***Long-Context Robustness Evaluation:*** The training conducted on 384-token sequences and the generalization of the resulting security policy for longer inputs remains an empirical question. The original SecAlign++ study addresses this for English-language data, but it has not been tested within the Vietnamese-dominant training configuration used here. A systematic evaluation across various input lengths (such as 512, 1,024, 2,048, and 4,096 tokens) using the existing fine-tuned model would help determine whether the short-context training regime generalizes to the longer contexts typical of production RAG and agentic deployments.

4.2.3 Long-Term / Advanced Directions

Three longer-term research directions are identified, each requiring methodological advances beyond the scope of the present study.

- 1) ***Online Reinforcement Learning for Securing Reasoning Models:*** The DPO-based approach used in this study is an offline preference optimization method, meaning that the preference dataset is constructed before training begins and does not change during the fine-tuning process. Recent research has shown that online reinforcement learning methods, particularly Group Relative Policy Optimization (*GRPO*; *DeepSeek-AI et al., 2026*), can yield stronger and more sample-efficient policy updates by adaptively generating preference data during training. Applying online reinforcement learning to prompt injection defense could involve designing a security reward function that penalizes responses that follow injections at each training step. This approach has the potential to enhance security with less training data. This direction is particularly relevant for reasoning-capable models, which tend to be more

vulnerable to prompt injection due to their enhanced ability to follow instructions and may benefit significantly from an adaptive training signal.

- 2) ***Visual Prompt Injection Defense for Multimodal Vietnamese Models:*** As the deployment of Vietnamese-language AI expands into multimodal applications (such as documents, forms, screenshots, and web interfaces), the potential for attacks also increases, extending beyond text to include image-embedded injections. In the text modality, the continuous nature of image representations makes adversarial perturbations much more potent than discrete token substitutions. Currently, there is no defense similar to SecAlign++ for visual prompt injection. Developing preference optimization-based defenses for multimodal models in the Vietnamese context would require a Vietnamese-language multimodal prompt injection dataset and a modified chat template that designates image inputs as untrusted data channels, similar to how text data is treated.
- 3) ***Persistent Memory Grafting Defense for Vietnamese Retrieval-Augmented Generation (RAG) Systems:*** In retrieval-augmented generation systems, the vector database storing retrieved document embeddings presents a persistent attack surface. An adversary who can inject documents into the retrieval corpus can influence all future model queries that access those documents without needing direct access to any single conversation. This type of attack, referred to as RAG poisoning in the OWASP Top 10 for LLM 2025, is not covered by the single-turn injection defense developed in this study. Creating a Persistent Memory Grafting defense for Vietnamese RAG systems would require both a threat model specific to the retrieval context and a training methodology that ensures the model resists instructions arriving through high-similarity document retrieval, rather than through direct data injection.

Chapter 5. Appendix

This appendix shows a comparative pipeline between the original research and this study, across data, processing, training, evaluation and result.

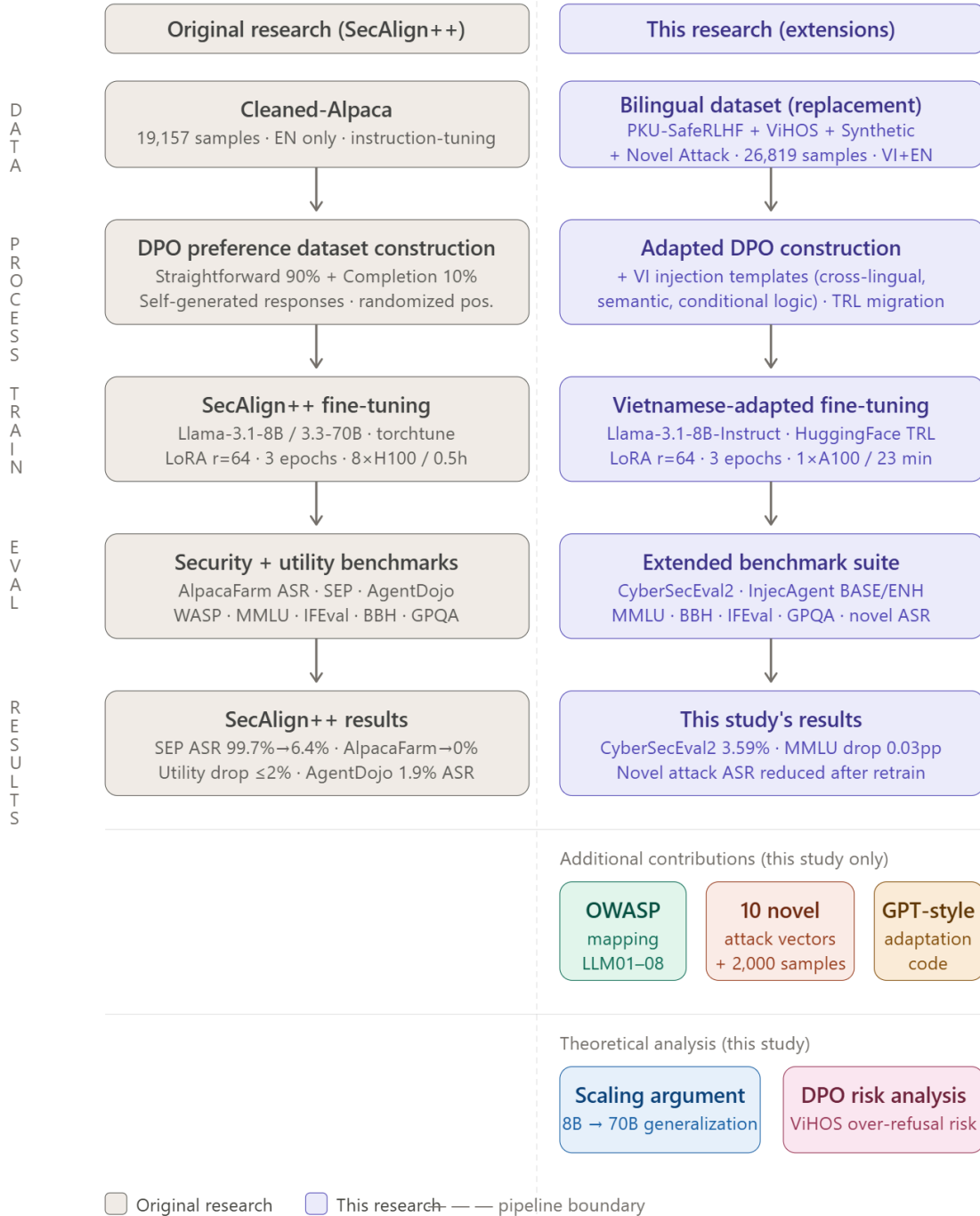


Figure 5.1. Comparative pipeline: SecAlign (original) vs. this study.

The left pipeline follows the original SecAlign++ recipe: a Cleaned-Alpaca English-only instruction-tuning dataset is used to construct DPO preference pairs via straightforward and completion attack simulation, fine-tuned on Llama-3.1-8B / 3.3-70B using torchtune across 8×H100 GPUs, and evaluated on AlpacaFarm, SEP, AgentDojo, WASP, and standard utility benchmarks.

The right pipeline extends SecAlign++ for the Vietnamese context across four stages. At the data stage, Cleaned-Alpaca is replaced by a 26,819-sample bilingual Vietnamese-English corpus combining PKU-SafeRLHF, ViHOS, a synthetic hallucination dataset, and a novel attack dataset. At the process stage, the DPO construction is augmented with Vietnamese-specific injection templates covering cross-lingual, semantic, and conditional logic attack patterns, and the training framework is migrated from torchtune to HuggingFace TRL. At the training stage, fine-tuning is completed on a single A100 GPU in 23 minutes with identical LoRA configuration ($r=64$, 3 epochs). At the evaluation stage, the benchmark suite is extended with CyberSecEval 2, InjecAgent BASE/ENHANCED, and a novel attack evaluation set covering all 10 identified vectors.

Beyond the core pipeline, this study contributes three additional components absent from the original work: a systematic mapping of SecAlign++ attack mechanisms to OWASP Top 10 for LLM Applications 2025 (LLM01-LLM08); empirical identification and targeted defense of 10 novel attack vectors with 2,000 generated preference samples; and a cross-architecture adaptation recipe enabling SecAlign++ deployment on GPT-style models. Two theoretical analyses are also provided: a scaling argument supporting generalization of the 8B security policy to 70B models, and a DPO risk assessment examining potential over-refusal induced by ViHOS-derived rejected responses.

ABBREVIATION

A	AI	Artificial Intelligence
	API	Application Programming Interface
	ASR	Attack Success Rate
B	BBH	Big-Bench Hard
	bf16	Brain Floating-Point 16-bit
C	ChatML	Chat Markup Language
D	DAN	Do Anything Now
	DPO	Direct Preference Optimization
G	GCG	Greedy Coordinate Gradient
	GPQA	Graduate-Level Google-Proof Q&A
	GPU	Graphics Processing Unit
	GPT	Generative Pre-trained Transformer
	GRPO	Group Relative Policy Optimization
I	IFEval	Instruction Following Evaluation
	ISE	Instructional Segment Embedding
L	LLM	Large Language Model
	LoRA	Low-Rank Adaptation
M	MMLU	Massive Multitask Language Understanding
N	NLP	Natural Language Processing
O	OWASP	Open Worldwide Application Security Project
P	PEFT	Parameter-Efficient Fine-Tuning
	PI	Prompt Injection
	pp	Percentage Points
R	RAG	Retrieval-Augmented Generation
	ROT13	Rotate by 13 Positions
S	SFT	Supervised Fine-Tuning
	SO	Specific Objective
T	TRL	Transformer Reinforcement Learning
V	ViHOS	Vietnamese Hate and Offensive Speech
	vLLM	Virtual Large Language Model

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